

Innovation Policy in Product-Line Markets: Evidence from GPUs*

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Abstract

I develop a dynamic oligopoly model to quantify the returns to R&D and the effects of innovation policies in product-line markets. In such markets, firms invest in R&D to advance the technological frontier and adjust their product menus to serve heterogeneous consumers and to compete with rivals. I apply the model to the desktop graphics processing unit (GPU) market, a duopoly between Nvidia and AMD. I find that, over 2011–2020, the average dollar of private R&D generates about \$9 of within-market surplus net of that dollar, with the firms capturing \$3 and consumers the remainder. I then compare the two levers of innovation policy: an R&D credit that lowers the cost of research, and product rebates intended to raise its reward. Counterfactual simulations reveal that public support generates far less surplus per dollar than the average social return to private R&D. A subsidy on frontier products can even discourage innovation: it may lower prices for consumers and raise the firms' profit, but reduce the incentive to advance the cutting edge.

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1 Introduction

When the GPU (graphics processing unit) was invented in the mid-1990s, it was a piece of computer hardware sold for one purpose: rendering better graphics in games. Three decades later, what were then circuit boards slotted into gamers' desktop computers are now the building blocks of modern data centers, stacked rack upon rack in warehouses, training frontier AI models. This leap did not happen overnight. It accumulated, one chip generation at a time, through continual R&D at two firms: Nvidia and AMD. Success stories like this, in which private R&D yielded sizable social value, provide the rationale for long-standing public policies designed to incentivize innovation.

Economists have long recognized the case for such policies: a firm cannot capture all the value its research creates, so the social value of research exceeds the private value (Arrow 1962). But two questions remain contested: how large the return is, and how the government should get involved. On the first question, the leading macro calculation (Jones and Summers 2022) puts the economy-wide social return to innovation at roughly \$5 to \$20, a fourfold span.¹ At the micro level, direct empirical evidence is scarce outside the pharmaceutical market (Bryan and Williams 2021). On the second question, governments now commit tens of billions of dollars to spur innovation using two levers:² *push* policies, such as R&D tax credits and US National Institutes of Health grants, lower the cost of doing the research; *pull* policies, such as advance purchase commitments and electric-vehicle subsidies, raise the reward for research whose output reaches the market. The heart of the matter is whether a public dollar buys anything at all rather than paying for what would have happened anyway, and, if so, which lever delivers the best bang for the buck.

I study both questions in vertically differentiated multiproduct markets³ using a dynamic oligopoly model that I apply to the desktop GPU market. I first estimate the market primitives. Based on those estimates, I derive the private returns and within-market social returns to the firms' R&D by running a counterfactual in which their R&D programs are shut down. I then simulate and compare three stylized policy interventions: a credit to the firms' R&D costs, a rebate on cutting-edge products, and a rebate on mass-market products. I report what a public dollar buys under each instrument and demonstrate how the interplay between pricing, product-line adjustment, and dynamic R&D incentives drives their effects apart.

¹This wide range is a result of an accounting exercise disciplined by growth theory, not by an econometric design: it prices the economy's productivity gains against its innovation spending, and the span reflects how broadly that spending is measured, how quickly benefits are assumed to arrive, and which benefits are counted.

²The United States CHIPS and Science Act of 2022 alone authorized \$52 billion for semiconductor manufacturing and R&D. Similar programs are in place in the European Union and across Asia.

³I refer to such markets as product-line markets throughout the paper.

Estimating the returns to R&D and comparing push against pull on equal terms require three things. The first is a cardinal measure of R&D output. In most empirical settings, research output is not observed directly; it is instead inferred from a proxy (e.g., productivity growth, patent counts, or citation counts), and the estimated return to R&D is only as good as the proxy.⁴ One exception is the pharmaceutical market, in which drug approvals and clinical trials give a direct count of the output of R&D, and researchers exploit exogenous variation in expected market size to identify how that output responds.⁵ The second requirement is a dollar-valued measure of welfare. Even in the pharmaceutical market, a count of new drugs is not welfare. Translating research output into dollars requires a model of consumers. A policy cost-benefit analysis further requires dividing the resulting surplus by the public dollars spent. Industrial organization studies routinely measure surplus through demand estimation and cost recovery. Few studies, however, derive surplus-based returns to R&D or measure the surplus a policy instrument generates per public dollar when firms' R&D spending and the technology frontier respond to the policy.⁶ Finally, the comparison requires a method for constructing the policy counterfactual. The reduced-form literature exploits exogenous variation to identify a policy's effect on measured outcomes, such as R&D spending, new products, or the movement of scientists across regions, but falls short where market power is first order⁷ and where natural experiments are lacking. The structural approach instead estimates primitives off other variation and solves the model again to recover the counterfactual outcomes. This approach carries its own costs, chiefly stronger assumptions about firm conduct.

Because welfare is hard to measure, policy debates often focus on a more tractable statistic: additionality, the amount of private R&D a public dollar crowds in. R&D spending is observed directly in financial reports, tax records, and R&D surveys, so additionality is easier to measure and far better documented: for the R&D tax credit, the most-studied instrument, leading reviews find that a public dollar induces roughly a dollar of additional private R&D.⁸ Governments use this statistic to certify programs, gauge value for money, rank instruments, and drive reform (e.g., [HM Treasury 2023](#)). Policy evaluations and the literature recognize that additionality is not a welfare

⁴The productivity branch of this tradition is laid out in [Griliches \(1979\)](#).

⁵[Acemoglu and Linn \(2004\)](#) use demographically driven variation in pharmaceutical market size to identify its effect on new-drug entry, and [Finkelstein \(2004\)](#) uses changes in vaccine policy to identify their effect on investment in clinical trials. This line of work is commonly referred to as the demand-pull literature.

⁶The closest work is [Goettler and Gordon \(2011\)](#), who study the CPU market. Their focus is on how competition affects innovation when products are durable and consumers are forward-looking. They use a single-product framework to maintain tractability but abstract away from the product-line structure, which is central to my work.

⁷One particular challenge is the risk of violating the stable unit treatment value assumption (SUTVA): subsidizing one firm induces its rival to reoptimize prices and qualities, so the rival is not an untreated control.

⁸Much of the classic evidence underlying the roughly dollar-for-dollar benchmark comes from cross-industry regressions of firms' R&D spending on its tax-adjusted cost ([Hall and Van Reenen 2000](#); [Hall 2022](#)).

measure, but how well the indicator tracks ultimate market outcomes remains unclear.

The desktop GPU market provides three advantages for studying these questions. First, the output of R&D is measured directly rather than via proxy: the quality of a chip is mainly reflected in its compute speed, which is tracked by standardized software. The same benchmark rates every GPU in the sample, so the cutting edge of 2020 can be compared to what was released in 2005 on a single metric. Second, I observe both firms' product lines over 16 years. Each firm maintains three tiers (low-end, midrange, and premium), and the panel records each tier's price, sales, and quality quarter by quarter, providing the variation needed to estimate demand and recover marginal costs. Third, the market structure held still while the technology raced. The same two firms competed throughout, with no entry or exit, and no major industrial policy targeted GPUs over the sample. The absence of change makes this market an ideal wind tunnel for studying the effects of a hypothetical policy intervention.

I build a model tailored to the institutional details of the GPU market: a dynamic oligopoly with endogenous pricing, quality positioning, and R&D. On the dynamic side, the firms track each other's technology frontier (i.e., the highest quality each can currently produce) and choose how much to spend on R&D. When R&D succeeds, the firm's frontier jumps, and, through a within-firm cost-spillover channel, the advance also lowers the production cost of the tiers below it. On the static side, the firms compete by choosing quality and price across the product line, taking into account both business-stealing and cannibalization effects. The policy comparison re-solves the estimated environment under each instrument.

I present two sets of results. The first is a pair of surplus-based private and social returns to firm R&D. Over 2011–2020, the average within-market social return is about \$9 per R&D dollar, while the industry net private return is about \$3.⁹ The wedge captures the surplus that accrues to consumers and that the firms cannot appropriate. Compared with the literature, the corresponding gross \$10 estimate sits at the lower end of the \$5 to \$20 that economy-wide growth accounting delivers: that range captures the whole economy's gains, while my measure counts only the surplus inside this one market. The pattern also matches the proxy tradition, which finds social returns of roughly 1.5 to 2.7 times the private return (Bloom et al. 2013). It is worth noting that neither match is built in: my structural approach shares almost no machinery with either literature. Rather, it reads output directly off the quality index and translates it into surplus through a partial-equilibrium model.

⁹Both returns are reported net of the R&D dollar itself, so zero is the break-even point: a dollar of R&D brings the industry about \$4 of gross profit and \$10 of gross surplus. The growth-accounting ratios cited in this section are gross, so the \$10 is the directly comparable figure.

The second result is a comparison of stylized push and pull policies under a common ex ante budget of \$5 billion over 10 years. On the push side, I apply an R&D credit that lowers the firms' cost of research, which in effect is a more aggressive and simplified version of the federal research and experimentation (R&E) tax credit the United States government has offered in some form since 1981.¹⁰ On the pull side, I consider two versions of a product rebate, both implemented as per-unit payments that lower a product's effective marginal cost. The premium rebate is paid at the cutting edge, but only on a small share of units sold. The non-premium rebate is paid on the mass-market models, aiming to expand overall adoption; it is the analog of an electric-vehicle purchase subsidy that restricts eligibility to models priced below a cap.¹¹ The fundamental difference between push and pull is who bears the risk: a push dollar goes out when the research money is spent, whether or not the research pans out; a pull dollar goes out only when the research output makes it to the market.

Counterfactual experiments reveal that public support returns far less surplus per dollar than private R&D does on average, despite the large social-private wedge. More counterintuitively, the premium rebate backfires on innovation: it crowds out R&D on net, returns about \$0.23 of within-market surplus per public dollar,¹² and leaves the end-of-horizon technology frontier about 0.3% below baseline for every \$1 billion of outlay. The R&D credit is the only instrument that induces a material increase in private R&D, and even its additionality is low: about \$0.13 of private R&D per public dollar. The credit returns about \$0.51 of surplus per public dollar and lifts the frontier by about 0.8% per \$1 billion. The non-premium rebate induces essentially no net R&D, yet it returns the most surplus—about \$1.09 per public dollar—and nudges the frontier up.

The backfire result, that subsidizing the frontier product can reduce the incentive to advance the frontier, arises from the interplay between the static incentive to maximize profit across the product menu and the dynamic incentive to innovate. A frontier advance does two things. At the top of the line, it improves the frontier product, and a better product commands a higher price. Below the top, knowledge gained during frontier research carries down to the production of the low-end and midrange tiers, saving cost on the units sold there. A rebate at the top expands the frontier product's sales, which on its own raises the value of an advance. But the rebate also sets off two forces that lower it. First, the product line sorts the least price-sensitive consumers to the

¹⁰The enacted credit is small in size and more intricate than a flat rate; for more details, see [Hall \(2022\)](#).

¹¹Electric-vehicle purchase subsidies are usually filed under environmental policy, but they are also innovation policy: they reward carmakers for improving the quality of electric vehicles, on which long-run adoption relies.

¹²As with the returns to R&D, zero is the break-even point.

top, so the new buyers are more price-sensitive than existing ones and value further quality less in dollars. Second, the rebate lowers the sales of the low-end and midrange tiers, so an advance saves production cost on fewer units. The net effect of subsidizing the top is therefore theoretically unsigned. Under the premium rebate I simulate, private R&D falls, even though the rebate raises industry profit by \$4.4 billion.

It is important to emphasize that the magnitudes of my empirical findings carry no external validity. The returns, the welfare signs, and the instrument ranking are specific to this market and the sample window. The counterfactual experiments evaluate stylized instruments that are chosen as stand-ins for real-world policies. I also focus on within-market effects: the surplus measure excludes downstream spillovers, such as those to the AI industry, as well as distributional, environmental, and general-equilibrium channels.

Nonetheless, at the conceptual level, three lessons may apply more broadly. First, in product-line markets with continual technological improvement, a subsidy for the frontier product can raise industry profit and still slow innovation: the product's profitability level and the incentive to improve its quality need not move together. The economic forces behind this backfire are present in other industries with a similar market structure, such as electric vehicles and consumer electronics, in which much innovation has taken place in recent years. Those forces, however, are much less salient in pharmaceutical markets, for which the empirical evidence on innovation policy is most abundant: a drug is rarely sold in quality tiers, and what it can treat is largely fixed at approval. Second, additionality is an incomplete policy yardstick. It tracks the R&D response but misses the response of prices, product choices, and adoption, so it can give a misleading picture of the surplus a public dollar buys. In my policy experiments, additionality is highest for the R&D credit, but the instrument that returns the most surplus per public dollar is the non-premium rebate, which expands access. Third, the social return and the social-private gap are both large in this industry, but neither magnitude implies that a public dollar returns what a dollar of firm R&D returns: in an innovation-driven industry, the private incentive could already be high enough that public dollars induce little R&D at the margin.

The rest of the paper is organized as follows. Section 2 provides institutional background on the market and describes the dataset. Section 3 builds the model. Section 4 presents the identification assumptions, the estimation procedure, and the parameter estimates. Section 5 measures the returns to firm R&D. Section 6 compares the instruments and discusses the policy implications. Section 7 concludes.

2 Background and Data

A modern personal computer runs on two types of processors. The CPU (central processing unit) is the computer’s brain: a general-purpose processor that works through tasks largely one after another and handles most routine computing, from filling in a spreadsheet to sending an email. In contrast, the GPU is built to run many simple calculations at once and handles a computer’s image and video rendering.¹³ Almost every computer ships with an integrated graphics processor, which is a basic GPU built into the CPU or the motherboard. Such processors typically provide enough processing power for routine work but not for heavy computation. Buyers who need more extensive processing power, particularly gamers, add discrete GPUs, which are standalone graphics cards, to their computers. I study the market for discrete GPUs from 2005 to 2020. During this period, the discrete GPU was mainly a consumer product, bought by gamers and by a small group of computer scientists who discovered that machine learning runs on the same parallel arithmetic. The AI boom that followed, most visibly after ChatGPT launched in late 2022, has shifted the GPU market toward business-facing demand concentrated among large enterprise buyers. That era falls outside the scope of this paper.

The desktop consumer discrete-GPU market over my sample period is a stable duopoly of Nvidia and AMD, whose GPUs jointly account for essentially all units sold.¹⁴ Both firms are so-called fabless designers: they design their chips and contract with outside foundries such as Taiwan Semiconductor Manufacturing Company Limited to fabricate them. Each firm maintains a product line that targets buyers across a wide range of willingness to pay. I observe unit sales, prices, and product quality for each firm at three tiers: low-end, midrange, and premium.¹⁵ Products arrive in successive generations that refresh the line.

These data form a balanced firm-tier-quarter panel: both firms offer all three tiers in each of the sample’s 64 quarters (2005Q1–2020Q4). Prices are inflation-adjusted, and every dollar figure reported from the data or the model is in 2018 dollars. Quality is measured by the PassMark G3D benchmark, a published performance test score. Each tier’s price and quality are averaged over the products the firm offers in that tier in the quarter. I observe each firm’s research spending in its quarterly reports and allocate it to the GPU segment in proportion to the segment’s share of

¹³If this page is read on a screen, a GPU drew it.

¹⁴The Radeon GPU line entered AMD through its 2006 acquisition of ATI; I treat the line as one continuing firm.

¹⁵Unit sales come from quarterly discrete-GPU shipment data purchased from Jon Peddie Research. The same source supplies the desktop CPU shipments that proxy for market size.

revenue.¹⁶ A potential buyer’s outside option is to add no discrete GPU and rely on the system’s built-in graphics. Every desktop that can take a discrete GPU has a CPU, so each desktop CPU sale counts as a potential buyer. I use those sales as the proxy for market size.

Table A.1 summarizes the product line, the market aggregates, and the measurement and sources of each series. Over the sample, benchmark quality at the frontier rose to about 100 times its 2005 level, while the premium-tier price roughly doubled. Figure A.1 plots the two series year by year. The tiers separate cleanly on both price and measured quality: the low-end and midrange tiers deliver a fraction of premium benchmark performance at a fraction of the price. The premium tier sets the performance frontier but accounts for only 7.6% of units sold. The sustained rise at the frontier is why the model in Section 3 treats the quality frontier as moving and R&D-driven rather than as a static dimension. Behind the moving frontier sits research spending: the two firms’ combined R&D runs at roughly a tenth of market revenue, \$1.2 billion against \$12.9 billion in an average year. Figure A.2 plots the by-firm series. Across the full sample, that spending totals about \$19 billion. Appendix A provides a fuller account of the market and these series.

The sharp contrast between quality growth and price growth implies that any measure of the return to innovation must account for quality change. If buyers value quality, a hundredfold quality gain sold at about twice the old price leaves much of the gain with them, and that portion never appears in the firms’ revenue reports. The demand estimates of Section 4 turn measured quality into willingness to pay. Section 5 then prices that gain in surplus terms.

3 Model

The model is a dynamic multiproduct oligopoly in the Ericson–Pakes tradition (Ericson and Pakes 1995), with endogenous product-line choice, Bertrand–Nash pricing, and endogenous R&D. Time t runs in quarters. Two firms, Nvidia and AMD, are indexed by f . Each firm maintains three tiers: low-end, midrange, and premium. A product is a firm-tier pair $j = (f, \text{tier})$, six inside products in all; $j = 0$ denotes the outside option.¹⁷

¹⁶Both firms report research spending as a single company-wide line: a segment split is not required by law, and neither firm provides one. Modeling competition in the firms’ other markets, in which each faces different rivals, is beyond the scope of this paper (see Rhodes et al. 2026). I adopt the proportional allocation as a simplifying assumption. The underlying economic assumption is that R&D spending per dollar of revenue is the same across a firm’s businesses. The GPU segment averages roughly 60% of Nvidia’s revenue over the sample and roughly a fifth of AMD’s. Appendix F.3 examines the sensitivity of the estimates and counterfactual results to this assumption.

¹⁷I do not model entry, exit, or adjustment in the number of tiers. The data come at the three-tier level, so the model is cast at the same level. Both firms offer the same three tiers in every sample quarter. Endogenizing the menu size would add substantial computational burden without corresponding insight into the research question.

The dynamic state is a five-dimensional vector, $x_t = (q_{1t}^{\max}, q_{2t}^{\max}, u_{0t}, z_t, M_t)$. The firm frontier $q_{ft}^{\max}$ is the highest quality firm f can currently produce. I refer to the firm with the higher frontier as the period's frontier leader and to the other as the follower. The roles switch endogenously as the state evolves. The outside-option level u_{0t} is the value of not buying a discrete GPU (using integrated graphics). The process-innovation state z_t is the industry-wide technical-progress index that lowers the cost of a given quality over time, reflecting the steady improvement of the foundries' manufacturing technology. Market size M_t counts the potential buyers. Of the five states, only the firm frontiers are endogenous, moved by the firms' R&D; the other three evolve exogenously.

The timing of the game is as follows. Each period opens with firms observing whether last period's R&D advanced their frontiers. Firms then simultaneously choose this period's R&D, which resolves at the next period's opening; this one-period lag is the standard time-to-build assumption of the dynamic-oligopoly literature (Ericson and Pakes 1995; Pakes and McGuire 1994). The within-period game follows: firms simultaneously choose the qualities of their low-end and midrange tiers, and then compete in prices à la Bertrand–Nash. I assume each firm sets the quality of its premium tier at its frontier; Section 3.2.1 specifies the two steps in detail and justifies this assumption. Consumers choose among the six inside products and the outside option, and demand is a random-coefficients model following Berry et al. (1995, henceforth BLP). Each firm maximizes the expected present value of its profits, discounted at the per-quarter factor β . Equilibrium is Markov-perfect. I now specify each block of the model in turn and discuss the underlying assumptions. Table G.1 collects the recurring symbols.

3.1 Demand

Demand is a standard BLP model.¹⁸ Each quarter, every potential buyer picks one of the six inside products or the outside option. Consumer i 's utilities are

$$u_{ijt} = \gamma q_{jt} + \alpha_i p_{jt} + \psi_{f(j)} + \xi_{jt} + \varepsilon_{ijt}, \quad j \in \mathcal{J}_t, \quad (3.1)$$

$$u_{i0t} = u_{0t} + \varepsilon_{i0t}, \quad (3.2)$$

where $\mathcal{J}_t$ collects the quarter's six inside products, q_{jt} is quality, p_{jt} is price, ξ_{jt} is unobserved quality, u_{0t} is the outside-option level, and ε_{ijt} is an i.i.d. type-I extreme-value taste shock. The

¹⁸The demand system is static and abstracts from forward-looking behavior. A full dynamic-consumer model is beyond the scope of this paper's research question. For the modeling of durable-goods demand, see Goettler and Gordon (2011).

brand fixed effect $\psi_{f(j)}$ captures brand loyalty and the software ecosystem specific to each firm. The single random coefficient is on price, $\alpha_i = \bar{\alpha} + \sigma_{\text{price}} v_i$ with $v_i \sim \mathcal{N}(0, 1)$.¹⁹ Integrating the choice probabilities over the price coefficient gives product j 's market share,

$$s_{jt} = \int \frac{\exp(\gamma q_{jt} + \alpha_i p_{jt} + \psi_{f(j)} + \xi_{jt})}{\exp(u_{0t}) + \sum_{k \in \mathcal{J}_t} \exp(\gamma q_{kt} + \alpha_i p_{kt} + \psi_{f(k)} + \xi_{kt})} d\Phi(v_i), \quad (3.3)$$

where Φ is the standard-normal distribution.

3.2 Supply

3.2.1 Stage Game

Within each quarter, the stage game is a two-step simultaneous-move game: firms first commit to the qualities of their interior tiers (the low-end and midrange), then set prices for all products. This timing assumption fits the industry: a fabless designer must contract with a foundry and commit in advance to the products it will manufacture, whereas retail prices can be adjusted flexibly within the quarter.²⁰ After qualities are committed, the demand and marginal-cost shocks, ξ_{jt} and $\varepsilon_{jt}^{\text{MC}}$, are realized, and the pricing step begins. The game is solved by backward induction.

I describe the two steps in reverse order. Pricing is a differentiated Bertrand–Nash game over all six products. The first-order condition for product j of firm f is

$$s_{jt} + \sum_{k \in \mathcal{J}_f} (p_{kt} - \text{MC}_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0, \quad j \in \mathcal{J}_f, \quad (3.4)$$

where $\mathcal{J}_f$ is firm f 's menu of three tiers and the marginal cost MC_{kt} is specified at the end of this subsection. Solving the six conditions jointly gives the equilibrium pricing map p^* . Each firm internalizes that a price cut at one tier draws demand from its own other two. Each product therefore carries the standard inverse-elasticity markup, widened where substitution runs within the firm. A frontier advance thus monetizes through the menu, not through one cell of it; it earns at the premium tier, shifts same-tier price competition, and cannibalizes the firm's own interior tiers. The rent from advancing the frontier is therefore itself a product-line object.

¹⁹Heterogeneity in the price coefficient alone provides the rationale for screening. The data do not provide enough variation to identify random coefficients on both quality and price precisely.

²⁰Quality-before-price timing is standard in the endogenous product choice literature (e.g., [Fan and Yang 2020](#)), and it leaves qualities predetermined relative to the period's demand and cost shocks. Section 4 justifies this assumption further based on institutional background and discusses the implications for identification and estimation.

At the quality step, each firm offers its premium tier at the frontier quality and holds its interior tiers at or below it. Equating the premium tier with the frontier is a simplifying assumption.²¹ The assumption makes the frontier an observed state, read off the premium benchmark, so the innovation process can be estimated from observed frontier advances rather than from a filtered latent state. The firm chooses its interior qualities to extract surplus from consumers with lower willingness to pay while protecting the rent its premium tier earns.

Formally, each firm chooses its two interior qualities, foreseeing the price equilibrium that will follow. It solves

$$\max_{\{q_{j't}\}_{j' \in \mathcal{J}_f^{\text{int}}}} \sum_{j \in \mathcal{J}_f} \mathbb{E}[\pi_{jt}(p^*(q), q)] \quad \text{subject to} \quad q_{j't} \leq q_{ft}^{\max} \text{ for } j' \in \mathcal{J}_f^{\text{int}}, \quad (3.5)$$

where $\mathcal{J}_f^{\text{int}}$ collects firm f 's two interior tiers, $\pi_{jt} = M_t s_{jt} (p_{jt} - \text{MC}_{jt})$ is product j 's variable profit, and the expectation is taken over the demand and marginal-cost shocks, neither yet realized when qualities are chosen. The first-order condition for interior product j' is

$$\sum_{j \in \mathcal{J}_f} \mathbb{E} \left[\frac{\partial \pi_{jt}}{\partial q_{j't}} + \sum_{k \in \mathcal{J}_t} \frac{\partial \pi_{kt}}{\partial p_{kt}} \frac{\partial p_{kt}^*}{\partial q_{j't}} \right] = 0, \quad j' \in \mathcal{J}_f^{\text{int}}. \quad (3.6)$$

The bracket adds a direct term, the profit response with prices held fixed, to an indirect term that works through the equilibrium price response of every product in the market, obtained from the pricing map p^* by the implicit function theorem. Raising a lower tier's quality wins some demand outright, but it raises the product's marginal cost, cannibalizes the firm's higher-priced tiers, and drags down equilibrium prices, including the rival's. The cannibalization is the own-product force distinctive to product-line competition (Fan and Yang 2020). The cost margin and the cannibalization together are why a firm holds its interior tiers strictly below the frontier.

The marginal cost of product j in quarter t is

$$\text{MC}_{jt} = c_0 + z_t q_{jt} + c_3 q_{jt}^2 + c_4 \left[\frac{q_{jt}}{q_{f(j)t}^{\max}} \right] + c_5 \left[\frac{q_{jt}}{q_{f(j)t}^{\max}} \right]^2 + \varepsilon_{jt}^{\text{MC}}. \quad (3.7)$$

The first three terms price the chip on its own: a constant, a linear quality term, and a quadratic quality term. The coefficient on the linear term is the process-innovation state z_t , equal to $c_1 + c_2 t$

²¹This assumption can be interpreted as follows: in equilibrium, the benefits of maintaining a gap between a firm's technological frontier and the highest quality it offers (a slack quality constraint for the premium product) do not outweigh the costs.

with $c_2 < 0$, so the cost of delivering a given quality falls over time. The decline captures advances in upstream foundry technology, exogenous to both firms, such as finer process nodes and rising yields as each node matures. The law of motion and long-run floor of z_t are specified in Section 3.2.2. The last two terms depend on $q_{jt}/q_{f(j)t}^{\max}$, the product's quality as a share of its firm's own frontier. This frontier-relative block captures design-technique reuse, a within-firm cost spillover: techniques developed at the cutting edge carry down the firm's own lineup.²² It gives the firm an additional incentive to advance its frontier: an advance lowers the cost of its interior tiers. This channel captures the intertwined process innovation and product innovation in this industry. Fixed costs are assumed constant in quality and so drop out of the first-order condition for quality. Section 4.2 carries the justification based on institutional background and the confirming evidence.

3.2.2 Dynamic Game

Over time, firms compete by choosing R&D spending. At each period's opening, firm f chooses an R&D intensity $\iota_{ft} \in [0, 1]$, its R&D spending as a share of expected per-period variable profit. Its value function on the state $x_t = (q_{1t}^{\max}, q_{2t}^{\max}, u_{0t}, z_t, M_t)$ solves

$$V_f(x_t) = \max_{\iota_{ft} \in [0,1]} \mathbb{E}[\pi_f^*(x_t)](1 - \chi \iota_{ft}) + \beta \mathbb{E}[V_f(x_{t+1}) | I_{ft}, x_t], \quad (3.8)$$

where $\mathbb{E}[\pi_f^*(x_t)]$ is expected equilibrium variable profit from the stage game. The firm's R&D spending is $I_{ft} = \iota_{ft} \mathbb{E}[\pi_f^*(x_t)]$, the dollars it books as R&D. The economic cost of that spending is χI_{ft} , where the cost scale χ converts each booked dollar into economic cost. A scale above one captures the economic costs that do not surface on the accountant's books: the managerial attention and organizational capacity a program absorbs, and the adjustment costs of assembling and holding a research team.

R&D carries an intrinsic risk of failure. The probability that firm f 's frontier advances at the next period's opening, conditional on its current frontier and its accounting spending, is

$$h_f(I_{ft}, q_{ft}^{\max}) = \frac{r_{ft}}{1 + r_{ft}}, \quad r_{ft} = \exp(a_f - \varphi q_{ft}^{\max}) g(I_{ft}), \quad (3.9)$$

where a_f is a firm-specific success-rate parameter, $\varphi > 0$ is the fishing-out parameter, and g is an increasing, concave function of R&D spending with $g(0) = 0$;²³ the saturating map $r/(1 + r)$ turns

²²The practice is documented on both sides of the duopoly (e.g., [Nvidia 2019](#); [AMD 2012](#)).

²³The baseline exposure is log-concave, $g(x) = \kappa^{-1} \log(1 + \kappa x)$, with curvature $\kappa = 0.5$. I report the sensitivity of the

the innovation rate into a probability, following [Pakes and McGuire \(1994\)](#). The hazard function h_f has three economic properties: zero R&D gives a zero advance probability; additional own R&D raises the probability at a diminishing rate; and, other things fixed, a higher current frontier lowers the advance probability, the restriction that $\varphi > 0$ imposes. Conditional on success, the size of the advance is an i.i.d. draw from a log-normal jump distribution. The hazard depends only on the firm's own R&D and its own frontier, with no cross-firm knowledge spillover. This exclusion fits the industry: a chip design is among a firm's most closely guarded secrets, and an advance by one firm does not readily travel to the other. The technical progress the two firms do share, the foundries' manufacturing improvement, is already in the model as the process-innovation state z_t .

The optimal intensity balances the cost and reward of research. Differentiating equation (3.8) with respect to ι_{ft} and dividing through by expected profit gives, at an interior optimum,

$$\chi = \beta \frac{\partial h_f}{\partial I_{ft}} \left(\mathbb{E}[V_f(x_{t+1}) | x_t, \text{advance}] - \mathbb{E}[V_f(x_{t+1}) | x_t, \text{no advance}] \right), \quad (3.10)$$

where the two expectations condition on whether firm f 's own frontier advances and integrate over everything else: the size of the jump and the rival's outcome. The condition prices the firm's trade-off: the left side is the marginal economic cost of an R&D dollar, and the right side is its discounted marginal benefit, the extra advance probability the dollar buys times the value of advancing over standing still. Because g is concave and the saturating map compresses high rates, the marginal benefit falls as spending rises, while the value gap does not depend on the spending itself. A higher cost scale therefore meets the falling benefit schedule at lower spending: given the continuation values, optimal intensity is weakly decreasing in the cost scale. This comparative static motivates using mean R&D intensity to estimate χ (Section 4.3). The scale is also the parameter the R&D-credit counterfactual acts on: a constant R&D tax credit is, in effect, a downward shift in χ .

The private optimality condition imposes no restriction on whether the industry underinvests or overinvests in R&D. Two forces pull the private reward away from the social value of an advance in opposite directions: the firm does not capture the consumer surplus its advances create, which depresses the private reward below the social value, and the business-stealing effect can lead to overinvestment, as a transfer between the firms creates no new value. The comparison is further complicated by the potential under- or over-provision of quality. Whether the equilibrium innovation rate sits above or below the socially optimal level is hence left to the data.

dynamic estimates to the hazard family and this curvature in Appendix [F.2](#).

The remaining three states lie largely outside the firms' control and evolve exogenously, each on a deterministic path disciplined by its own in-sample dynamics. The outside option improves over time but not without limit: its level drifts upward along a linear trend toward a long-run ceiling,

$$u_{0,t+1} = \min\{u_{0t} + \Delta_u, u_{0,\max}\}, \quad (3.11)$$

where the per-period drift Δ_u is estimated from the recovered demand time effects and the ceiling $u_{0,\max}$ is a closure object beyond the sample. The process-innovation state, the technical-progress index in marginal cost (equation (3.7)), falls as manufacturing technology advances on a publicly telegraphed industry roadmap, again not without limit: it drifts downward toward a long-run floor,

$$z_{t+1} = \max\{z_t + c_2, z_{\min}\}, \quad (3.12)$$

where $c_2 < 0$ is the in-sample quality-by-time cost trend (from equation (3.7)) and the floor $z_{\min}$ is again a closure object.²⁴

Market size mean-reverts in logs toward a long-run level,

$$\log M_{t+1} = \log M_\infty + \rho_M(\log M_t - \log M_\infty), \quad (3.13)$$

with the long-run level M_∞ and the persistence $\rho_M \in (0, 1)$ calibrated to the late-sample level and persistence of the observed series.

I focus on stationary Markov-perfect equilibria of the game: policies and value functions depend on the state alone, not on calendar time. The state also stays effectively bounded. The exogenous laws of motion impose their long-run bounds directly, while frontier growth is restrained by diminishing innovation incentives: advances from a high frontier become both harder to draw and worth too little additional product-market profit to justify the spending, so R&D tapers off. For the Ericson–Pakes class in general, existence of a pure-strategy Markov-perfect equilibrium is not guaranteed, and equilibria need not be unique (Doraszelski and Pakes 2007; Doraszelski and Satterthwaite 2010). As is standard practice in this literature, I assume the data are generated by a single equilibrium and solve for it.

²⁴A closure bound is needed for two reasons: realism, since neither the outside option nor manufacturing technology improves without limit, and tractability, since bounding the exogenous states keeps the stationary dynamic game solvable. In the baseline, both trends flatten at their bounds about five years past the estimation sample, well beyond the horizon over which I score the simulations (Section 4.2), so the closure does not bind on the simulated paths. Appendix D.1 makes this precise, and I report sensitivity to later plateau dates in Appendix F.2.

4 Identification and Estimation

There are four blocks of primitives to be estimated: demand, marginal cost, the transition process, and the R&D-cost scale. For each block, I state the identification assumptions and estimation procedure, then report the estimate and model fit. One notable feature of the methodology is the neural-network surrogate that approximates the within-period equilibrium. Readers primarily interested in the policy results can skip ahead to Sections 5 and 6 without loss of continuity.

4.1 Demand

The timing of the stage game makes price endogenous in the demand system, a familiar problem in BLP-style estimation: qualities are committed before the period’s demand shocks are realized, while prices are set afterward. Price therefore co-moves with the demand unobservable ξ_{jt} .

I address the endogeneity issue with price instruments. The lead instrument is a shift-share-style IV built on the semiconductor components producer price index (PPI), lagged one quarter,²⁵ interacted with the premium and midrange tier indicators, with the low-end tier omitted as the base. Because the index is a single quarterly time series, it is absorbed by the quarter fixed effects. The identifying variation therefore comes from the tiers’ differential exposure to input-cost shocks: high-end chips use larger dies with lower fabrication yields, so their marginal cost is more responsive to input costs than that of low-end chips. I assume the input prices are set upstream in the global silicon market, orthogonal to the period’s GPU demand shocks.

I supplement the cost shifter IV with two families of BLP-style instruments: the Gandhi–Houde quadratic differentiation instruments in own and rival quality (Gandhi and Houde 2019), constructed with the routines of Conlon and Gortmaker (2020) as sums of squared quality distances to the firm’s other products and to rival products; and a supplementary moment in the square root of own quality. Both families are functions of product characteristics predetermined before the period’s demand shock is realized.²⁶ The demand first stage is strong, with a joint F statistic of about 49. Appendix B.1 reports the full first-stage regression table.

Table 1 reports the estimates: the parameters in Panel A, with a positive quality coefficient and

²⁵The BLS series covers transistors, diodes, and semiconductor parts such as wafers.

²⁶I assume that no part of the current shock is forecastable when the menu is set, so these functions are orthogonal to ξ_{jt} , unlike the price they instrument for; this is the standard basis for instrumenting price with functions of predetermined characteristics (Eizenberg 2014). The square-root own-quality term carries one further requirement: as a nonlinear function of a characteristic that itself enters utility, it is a valid excluded instrument only under the linear-in-quality index, which keeps it out of the demand equation. It enters as a supplementary moment. Conlon and Gortmaker (2020) use analogous moments.

a negative price coefficient, both estimated precisely; the implied own-price elasticities in Panel B; and the markups those elasticities imply through pricing optimality in Panel C. The elasticities rise across tiers, about 1.94 at the low-end tier, 3.29 at the midrange, and 4.41 at the premium. Because utility is linear in price, the own-price elasticity rises with the price level: on a more expensive product, a given percentage price change is a larger dollar amount, and is hence a larger utility loss. The premium tier, priced highest, faces the most elastic demand even though premium buyers are the least price-sensitive: the price-level effect outweighs their lower sensitivity. Table B.2 reports the full elasticity matrix. The estimate of the random coefficient on price (σ_{price}) shows a high level of consumer heterogeneity, which sorts the least price-sensitive, highest-willingness-to-pay buyers toward the premium tier and the most price-sensitive toward the low-end tier or the outside option.

The dollar markup and the Lerner index then measure market power in two ways. In dollars, the markup rises across tiers, from about \$80 to \$100 to \$165. The markup ratio falls across tiers, from 0.61 to 0.37 to 0.29, because price climbs faster than the markup. The premium tier thus carries the largest absolute markup but the thinnest rate: the pattern reverses the presumption that luxury goods carry the fattest markup rates, but it is a reasonable outcome in this market. I report sensitivity analysis across sample range and market-size definition in Table F.1.

The outside-option value is backed out from the quarter fixed effects; it rises roughly linearly, by about \$6 per quarter, as integrated graphics improve (Figure D.1), and enters the dynamic model as an evolving exogenous state (Section 4.3.1). Together, the included variables explain about 73% of the variation in the mean utilities recovered from the standard Berry inversion procedure (Berry et al. 1995).

4.2 Static Cost Primitives

Marginal costs are recovered from the standard pricing first-order conditions. I then estimate the coefficients of the marginal-cost function by OLS on the recovered series. The stage game's timing means no instrument is needed at this step: qualities are chosen and the frontier is realized before the period's cost shock is drawn,²⁷ and I assume the cost shocks are serially uncorrelated, so the regressors are predetermined with respect to that shock. Table 2 reports the coefficients. Marginal cost is convex in own quality, consistent with the falling fabrication yields of the larger dies that higher quality requires. The process-innovation trend enters with a negative slope: delivering a

²⁷The timing fits the industry. Committing a quality binds the design to a foundry process and to a supply chain that must be coordinated far in advance. The period's cost shock, in contrast, reflects input prices and fabrication yields, neither of which is settled until well after the design is frozen.

Table 1. Demand estimates

<i>Panel A. Parameter estimates</i>					
Price				−17.83	(5.30)
Quality				60.51	(12.01)
Random coefficient on price (σ_{price})				5.63	(2.87)
Firm fixed effects					Yes
Quarter fixed effects					Yes
Observations (firm-tier-quarter)					384
<i>Panel B. Own-price elasticities</i>					
Low-end					1.94
Midrange					3.29
Premium					4.41
<i>Panel C. Markups and marginal cost (low-end / midrange / premium)</i>					
Average price	\$132	/	\$267	/	\$570
Recovered marginal cost	\$52	/	\$167	/	\$405
Markup	\$80	/	\$100	/	\$165
Lerner index (markup ratio)	0.61	/	0.37	/	0.29

Notes: Two-step GMM estimates of the demand system of Section 3.1 with a single random coefficient on price; instruments are the tier-interacted cost shifter, the quadratic Gandhi–Houde differentiation instruments, and the square-root own-quality moment. Heteroskedasticity-robust, unclustered standard errors are in parentheses; sample period 2005–2020. Price enters in thousands of dollars and quality in units of 100,000 benchmark points, so the Panel A coefficients are on those scales; the Panel B elasticities are scale-free. Panel B averages the absolute own-price elasticities across markets within each tier; the all-product mean is 3.21. Panel C markups follow from the estimated elasticities through the pricing first-order conditions. Price, marginal cost, and markup are unweighted tier means; the Lerner index is tier-mean markup over tier-mean price. Recovered marginal cost is positive for all but four of 384 observations. Appendix B.3 reports prices, marginal costs, and markups by firm and tier.

Table 2. Marginal-cost estimates

	Estimate (SE)
Constant (c_0)	0.05 (0.02)
Quality (c_1)	1.07 (0.41)
Quality $\times$ time, process trend (c_2)	-0.04 (0.01)
Quality squared (c_3)	18.18 (1.35)
Relative quality, $q/q^{\max}$ (c_4)	-0.08 (0.07)
Relative quality squared (c_5)	0.30 (0.05)
R^2	0.87
Observations (firm-tier-quarter)	384

Notes: Ordinary least squares on the marginal-cost series recovered from the pricing first-order conditions; standard errors in parentheses. All parameters except the linear relative-quality term (c_4) are significant at the 0.01 level. Statistical significance is calculated using unrounded estimates.

given quality gets cheaper as upstream manufacturing technology improves. The frontier-relative terms explain a nontrivial share of the variation in marginal cost: dropping them lowers the R^2 by 0.15. The data thus support the within-firm cost spillover of Section 3.2.1. Overall, this rich specification fits the recovered series closely, with an R^2 of 0.87.

The coefficients for the frontier-relative terms carry an incentive implication: a given interior quality becomes cheaper to build as the firm's frontier advances. What a firm learns in pushing its frontier is not specific to the chip that first embodies it. Design techniques first developed for cutting-edge models are reused across the firm's own lineup, so an advance lowers the marginal cost of the interior tiers rather than upgrading the flagship alone.²⁸ The interior tiers therefore give the firm a stake in advancing its frontier.

Manufacturing high-end chips requires expensive capital investment. I nonetheless treat the fixed cost as constant in quality, so it drops out of the firm's quality-choice problem. In reality, the fixed cost is largely a one-time expenditure: chips of the same generation share the same basic architecture, and the expense of designing that architecture is incurred before the generation's tier qualities are set, so it is invariant to them.²⁹ The larger platform costs, such as the move

²⁸In the model, the frontier-relative block of equation (3.7) is $c_4 (q_{jt}/q_{f(j)t}^{\max}) + c_5 (q_{jt}/q_{f(j)t}^{\max})^2$. Differentiating marginal cost with respect to the firm's frontier $q_{f(j)t}^{\max}$, holding q_{jt} fixed, gives $\partial MC_{jt}/\partial q_{f(j)t}^{\max} = -(q_{jt}/(q_{f(j)t}^{\max})^2) [c_4 + 2c_5 q_{jt}/q_{f(j)t}^{\max}]$, which is negative whenever $q_{jt}/q_{f(j)t}^{\max} > -c_4/(2c_5)$: an advance lowers the marginal cost of every product built above that ratio. Computed from the unrounded coefficient estimates, the threshold ratio is roughly 13%. On the estimation panel, every midrange observation lies above it, as do most low-end observations.

²⁹The industry refers to this one-time cost as non-recurring engineering (NRE).

to extreme-ultraviolet lithography, are borne further upstream: by the foundries and by their equipment suppliers, chief among them ASML, rather than by the fabless designer. They also arrive on an update cycle of their own. I thus assume that repositioning an interior tier’s quality reuses the same design and the same manufacturing process, so the right-hand side of equation (3.6) is zero rather than a function of quality, the more typical specification in the endogenous product choice literature (Fan 2013).

The data support the assumption. At the observed qualities, the left-hand side of equation (3.6) can be computed from the estimates: the direct profit derivatives come from the demand system and the cost function, and the equilibrium price response comes from the implicit function theorem applied to the pricing first-order conditions. I evaluate it at every firm-quarter for the two interior tiers, integrating over the demand and cost shocks, following the total-differentiation approach of Fan (2013). Interpreted as an implied marginal fixed cost of quality, the object is large and swings widely in the early years; it falls by nearly two orders of magnitude across the sample and sits near zero over the post-2010 window (Figure B.1). I read the early years as an emerging industry not yet settled into stable quality-setting, and 2011 onward as the stationary regime the dynamic model describes. The static primitives, demand and marginal cost, are recovered on the full sample. The R&D-cost parameter and the counterfactuals, which both rely on the quality first-order condition, use the post-2010 window, 2011Q1–2020Q4. This window also sets the paper’s scoring horizon: every counterfactual object is computed over 10 years, starting from the initial state of the evaluation period.

4.3 Dynamic Estimation

The dynamic block has two kinds of primitives: the laws of motion for the state variables, and the R&D-cost scale χ , the economic cost of the research activity. As is standard in dynamic-games estimation, I estimate the transition parameters offline, outside the game: those parameters are identified from the functional-form assumptions and the observed state transitions, with no equilibrium computation needed. I use mean R&D intensity to estimate the cost scale. The optimality condition in Section 3.2.2 motivates this moment: holding the state and continuation values fixed, a higher χ raises the marginal cost of R&D and weakly lowers optimal intensity. I estimate the scale by full solution, re-solving the dynamic game at each trial value and matching its simulated mean intensity to the data. With one moment for one parameter, the estimator is equivalent to a root-finding exercise and is exactly identified. With two firms and one market, a full

solution is the only practical way to estimate the dynamic game, as the data are not rich enough for two-step methods such as [Bajari et al. \(2007\)](#). Overall, the dynamic block is less tightly disciplined by the data than the static primitives, an intrinsic limitation of the single-market setting. I therefore keep the dynamic model and its estimation parsimonious,³⁰ and report sensitivity analysis in Appendix [F.2](#).

4.3.1 The Transition Process: Frontier Dynamics and Exogenous States

The innovation hazard (equation (3.9)) is estimated by maximum likelihood on the innovation-success panel: each firm’s record of frontier advances over the sample period, paired with the R&D spending and the current frontier level. The estimation recovers the two firm-specific success-rate parameters and the common fishing-out parameter. The conditional frontier-jump distribution has its log-mean and log-variance fixed at their sample moments. The hazard is estimated using the full history rather than the shorter post-2010 window.³¹

The three exogenous states are estimated as deterministic processes, each disciplined by its own in-sample dynamics. The outside-option level is recovered by inverting the quarter fixed effects (Section 4.1) and converted to dollars with the price coefficient. Because the benchmark measures product quality directly, the quarter effects load onto the outside option rather than onto unmeasured quality drift in the inside goods. The fitted trend runs from roughly $-\$120$ in 2005 to roughly $\$260$ in 2020. The recovered series shows a dip in 2017–2018 and then returns quickly to its overall path (Figure [D.1](#)). This dip likely picks up the crypto-mining boom of those quarters: mining then ran on GPUs, and the surge in prices and volumes is rationalized by the demand system as an outside option that is temporarily much worse. I exclude those quarters when fitting the trend.

The process-innovation state needs no separate estimation: z_t is the time trend already estimated in the marginal-cost function, $z_t = c_1 + c_2 t$ (Table 2). Market size over my sample period shows the pattern of a maturing technology: desktop CPU sales fall through the middle years and level off toward the end as adoption saturates. I model market size as mean-reverting in logs toward a long-run level, calibrated on the post-2010 window; like the other states, it enters the dynamic model as a deterministic fitted path rather than as stochastic draws.

³⁰Empirical work on dynamic games often estimates several parameters from many moments. In my setup, only one parameter, χ , is estimated inside the game, and its identification logic is more straightforward than is typical in the literature. Additional moments would add little information about χ , so I keep the estimator exactly identified.

³¹The post-2010 window of Section 4.2 guards the quality first-order condition, whose implied fixed cost settles only after 2010; the transition estimates never use that condition, so the full history is available to them.

4.3.2 The Static-Equilibrium Surrogate

The full-solution estimator re-solves the entire dynamic game at every trial value of χ . The quality-then-price stage game inside each period compounds the burden: every update of the value functions needs that game’s equilibrium at a large number of states, and solving it exactly at each one is computationally prohibitive. An exact solve is a nested pair of fixed points: a quality equilibrium computed by best-response iteration, each step of which re-solves the six-product pricing equilibrium at every draw of the period’s demand and cost shocks.

I approximate the stage game with two nested neural networks, a pricing surrogate and a quality surrogate. The pricing surrogate maps the six product qualities, the outside-option level, and the process-innovation index to the shock-integrated outcomes of the pricing equilibrium: expected prices, shares, variable profits, and consumer surplus. The quality surrogate maps the two frontiers and the same two exogenous states to the equilibrium interior qualities. Market size enters neither network: it scales profits and consumer surplus without moving equilibrium prices or qualities, so it multiplies the surrogate outputs directly. Training mirrors the nesting, in two passes: I solve the pricing game exactly on a large sample of quality menus and states and fit the pricing surrogate to those solutions. Then, with the fitted pricing surrogate as the inner map, I solve the quality game across sampled states and fit the quality surrogate to the result. For each surrogate, I train 20 independent networks using different random seeds and use their equal-weight ensemble as the final model.³²

The training draws span the historical range and a wider region beyond it, and include the observed post-2010 states and small perturbations around each. For the policy exercises of Section 6 that change the stage game, I resimulate and refit the surrogates, appending to the training set the states visited by the estimated equilibrium’s simulated paths, so each refit trains densely where the policy paths travel. Appendix C details the construction and training. Approximating equilibrium objects and decision rules with neural networks is a standard tool in high-dimensional dynamic programming and reinforcement learning, but remains less common in empirical dynamic games.³³

The surrogates fit well. Held-out R^2 is 0.99 for the pricing and 0.97 for the quality game, and the networks improve on polynomial and LASSO benchmarks (Table C.1), with the largest gains on shares and profits, the objects the surplus measures are built from. I choose the training bounds so

³²Unlike a regression, training a neural network carries intrinsic randomness; averaging independent runs is insurance against an unrepresentative draw.

³³Bod  r   (2026) similarly uses a neural network to approximate stage-game equilibrium profits and to construct basis functions for the value function in a full-solution empirical dynamic game.

that state visitation in the dynamic estimation and the counterfactuals is confined to the training region, with no extrapolation beyond the bounds.

4.3.3 Solving the Game and Estimating the Cost Scale

At each trial value of the R&D-cost scale, I solve the dynamic game by iterative best response: I initialize both firms' policies at zero investment, compute each firm's value function and R&D policy as a best response to the rival's, and iterate the pair to a fixed point. The dynamic game has a five-dimensional continuous state space and a continuous control, which rules out an exact solution; I thus approximate the value function on a Chebyshev basis.³⁴ At each state the solver visits, the surrogates supply the period payoff, the expected equilibrium variable profit $\mathbb{E}[\pi_f^*(x_t)]$. Appendix D.2 details the solution method and the simulation protocol.

The cost scale is estimated by the method of simulated moments: with the game solved at a trial value, I simulate the model over the post-2010 window, 2011Q1–2020Q4, compute its mean R&D intensity, pooled across firms and quarters, and adjust the scale until the simulated mean matches the sample value. The observed intensity mirrors the model's definition (each firm's R&D spending divided by the expected variable profit that the estimated static primitives imply at the observed frontiers, with the exogenous states on their fitted paths) and averages about 21%. Because the parameter is exactly identified, I match this moment exactly, up to solver tolerance. Appendix D.3 states the estimator formally.

4.3.4 Estimation Results and Model Fit

Table D.1 reports all the parameters of the dynamic block. Panel A carries the main parameter, and the only one estimated inside the dynamic game: the R&D-cost scale χ is 5.13. The scale reads as an economic cost of about 5.1 times the firm's accounting R&D cost. The excess over the accounting outlay is a familiar object in the user-cost-of-R&D literature, in which it appears as the marginal adjustment cost of R&D (Hall and Lerner 2010), a term posited but rarely measured directly. I read the parameter estimate as a self-consistent, reduced-form measure of that user cost, implied by the firms' observed intensity. It should not be read as a direct measure of the cost of research. Figure D.2 plots the estimation objective at the evaluated values of χ , with the lowest value at the estimate.

³⁴The parametric value-function approximation follows Sweeting (2013).

Panels B and D collect the offline blocks of Section 4.3.1: Panel B the frontier transition (the hazard’s by-firm success-rate parameters, the common fishing-out parameter, and the frontier-jump distribution matched to its sample log-moments); and Panel D the fitted exogenous-state processes (the outside-option trend, the process-innovation state, and the market-size law of motion). Across both panels, the estimates are of economically sensible magnitude. Within Panel B, the firm-specific success-rate parameters are estimated precisely, while the fishing-out parameter is not, a consequence of the rarity of frontier advances.³⁵ The discount factor β is fixed at 0.99 per quarter.

Despite having only one free dynamic parameter, matched to a single pooled moment (the mean R&D intensity), the model holds up well beyond that moment. I simulate the estimated equilibrium and plot the 10th-to-90th percentile bands of its by-firm paths of market share, frontier quality, and R&D intensity, layering on top the paths at the observed state path (Figure D.3). None of these paths is targeted in the estimation, yet they sit largely within the bands. The exceptions mostly fall in the final years, when both firms’ R&D intensities climb above the bands, most persistently for Nvidia, whose frontier runs above as well.³⁶ With the primitives in hand, I simulate the market under counterfactual restrictions and policy schemes.

5 Returns to Firm R&D

I open the counterfactual analysis with several measurements, namely, how much surplus a dollar of a firm’s own R&D generates, how much profit it creates, and the wedge between the two.

The return to a firm’s R&D is measured against the world in which that R&D does not happen. I set each firm’s R&D to zero permanently, resimulate the equilibrium path, and evaluate the effects over the 10-year window (2011–2020) on which the dynamic estimation is based.³⁷ The expected difference from the estimated equilibrium is what the firm’s R&D produces, for the firm in operating profit and for the market in within-market surplus. The shutdown runs under two conventions. In the by-firm experiments, the rival keeps its estimated-equilibrium R&D policy rule, although its realized spending still changes because the state path differs. Each by-firm experiment

³⁵With a small sample, theoretically motivated parametric assumptions naturally play a larger role in disciplining the estimation. I therefore reestimate the R&D-cost scale under alternative assumptions about the hazard and the other primitives of the dynamic specification; Table F.2 reports the results. Across the variants estimated, the scale ranges from about 3.2 to 5.5, and the reading of an economic cost well above the accounting cost holds in every case.

³⁶One potential reading of Nvidia’s late-sample surge is R&D aimed at demand outside this market: if the firm was already positioning for the boom in AI accelerators, which arrived after the sample ended, an R&D program sized to that opportunity would rationally exceed what desktop-GPU profits alone justify, and the model prices only the latter.

³⁷The market has shifted drastically since the sample ended, most visibly with the AI boom, so I evaluate all counterfactual results within this window rather than as full discounted payoffs.

therefore measures the effect of a unilateral deviation in which the shutdown firm sets R&D to zero. In the full-industry experiment, both firms' R&D shuts down at once, so neither firm falls behind and the competitive pressure to innovate vanishes. In both conventions, the exogenous state variables evolve as in the estimated environment. Appendix E.1 carries the full counterfactual protocol. Throughout, I focus on within-market surplus as the conservative policy-evaluation object: it captures the benefits to discrete-GPU consumers, and it excludes the downstream and user spillovers (e.g., to AI, cloud computing, autonomous vehicles, or national security).

I report each return as a ratio of present values: the net private return is the change in the firm's own operating profit per dollar of its own R&D, net of that dollar; the social return is the change in within-market surplus³⁸ per that same dollar; and the social-private wedge is the difference between the two. Formally, for the shutdown of firm f 's R&D, write Δx_{t+h} for the equilibrium-minus-shutdown change in a value x in quarter $t + h$ of the horizon. With sums discounted at the per-quarter factor β over $h = 0, \dots, 39$, the two returns are

$$\text{net private return} = \frac{\sum_{h=0}^{39} \beta^h \Delta \Pi_{f,t+h}}{\sum_{h=0}^{39} \beta^h I_{f,t+h}} - 1, \quad \text{social return} = \frac{\sum_{h=0}^{39} \beta^h \Delta W_{t+h}}{\sum_{h=0}^{39} \beta^h I_{f,t+h}},$$

where the minus one nets out the R&D dollar itself, because operating profit is measured before R&D spending; the social return needs no such adjustment, because W already deducts the R&D spending I . Each denominator is the present value of the firm's own R&D program: its spending $I_{f,t+h}$ along the estimated-equilibrium path (the shutdown path spends nothing). The full-industry variant applies the same construction with both firms' R&D shut down and the industry's discounted R&D in the denominator. These returns are program-level averages over the firm's entire R&D, reported as dollar-denominated ratios of present values: the average return to the whole program, not the marginal effect of an additional R&D dollar.³⁹ All returns I report are expected values: averages over simulated paths that all start from the 2011Q1 state.

³⁸Within-market surplus $W = CS + \Pi - I - G$ deducts the firms' accounting R&D spending I and the public outlay G from consumer surplus plus the firms' profit Π , which adds any public transfers the firms receive to their operating profit measured before R&D. I keep the accounting convention as the baseline because it is more comparable to the broader innovation literature, which focuses largely on reported R&D. The R&D deduction is deliberately smaller than the cost χI the firm bears in its dynamic problem (equation (3.8)): the excess over the booked outlay is not a recorded expenditure, and how much of it is real resource cost rather than rents in the labor market is not separately identified, so how much of it to deduct from surplus has no unique answer. Under the polar alternative that treats that full excess as real resource cost, both returns fall in level, the industry wedge is exactly unchanged, and the by-firm wedges widen. Absent a policy, as in this section, the transfers and G are zero.

³⁹The marginal return is the harder object to define and interpret in a dynamic framework: the effect of an additional R&D dollar depends on when the increment happens and how spending adjusts around it, so each timing convention defines a different experiment. The program average requires no such convention.

Table 3. Returns to firm R&D

R&D shut down	Present-value dollars per dollar of own R&D		
	Net private return	Social return	Social-private wedge
AMD	3.33	6.00	2.67
Nvidia	3.81	7.57	3.76
Full industry	3.00	8.95	5.95

Notes: Each entry is a ratio of present values: the equilibrium-minus-shutdown change in the numerator, the discounted accounting R&D that the shutdown removes in the denominator. The net private return is the change in the firm's operating profit per dollar of its R&D, net of that dollar; the social return is the change in within-market surplus, which deducts the firms' accounting R&D and excludes downstream spillovers; the social-private wedge is the social minus the net private return. Each by-firm row shuts down the named firm's R&D while the rival keeps the policy rule it follows in the estimated equilibrium; the full-industry row shuts down both firms' R&D at once and divides by the industry's discounted R&D. Present values run over the 10-year horizon 2011–2020, 40 quarters, discounted at $\beta = 0.99$ per quarter. For scale, the full-industry shutdown removes about \$7 billion of discounted R&D and with it about \$64 billion of within-market surplus. Appendix F.2 repeats the full-industry experiment under alternative dynamic specifications.

Table 3 reports the net private and social returns to firm R&D, and the wedge between them, by firm and for the full industry. The social return exceeds the net private return in every row. A dollar of AMD's R&D returns \$3.33 to the firm and \$6.00 to the market, a wedge of \$2.67; a dollar of Nvidia's returns \$3.81 and \$7.57, a wedge of \$3.76.⁴⁰ The full-industry row carries the lowest net private return in the table, \$3.00, the highest social return, \$8.95, and the largest wedge, \$5.95 per R&D dollar. In present-value terms, the full-industry shutdown removes roughly \$7 billion of discounted R&D and, with it, roughly \$64 billion of within-market surplus, about half the undiscounted revenue the market realized over the decade.

The full-industry net private return, about \$3 per R&D dollar, lies about \$0.3 to \$0.8 below the by-firm private values. The gap captures business stealing: in the by-firm rows, part of what a firm's R&D earns is the shares and margins defended against a rival that keeps innovating. When both firms stop at once, neither falls behind and this component nets out of the industry aggregate. Conceptually, the effect is similar to the rent dissipation of a patent race (Loury 1979). The social side runs the other way, and more strongly: the full-industry social return, \$8.95, exceeds the returns in the by-firm rows by roughly \$1.4 to \$3.0 per R&D dollar. The simultaneous shutdown removes about \$14.6 billion more within-market surplus than the two unilateral shutdowns combined. More

⁴⁰The by-firm rows hold the rival to its estimated-equilibrium policy rule. In a companion exercise, the rival instead reoptimizes its R&D policy against the permanent shutdown. The private-return rows change by less than \$0.03; the social-return responses are larger and asymmetric: AMD's rises by about \$0.68, while Nvidia's falls by about \$0.17.

broadly, with market power, the return to R&D is an equilibrium object: its definition requires an assumption about rival behavior.

It is informative to put my numbers beside the estimates from the literature. Returns to R&D measured from estimated demand are rare.⁴¹ The broader economics of innovation literature nonetheless offers magnitudes close enough to compare. The first comparison is the level of the social return. [Jones and Summers \(2022\)](#) put the social value of innovation investment, economy-wide and inclusive of cross-market spillover channels, at roughly \$5 to \$20 per dollar invested; against the narrower base of R&D spending alone, their estimate is about \$13. The \$8.95 here is within-market surplus per dollar of the industry's own R&D, net of that dollar (gross, \$9.95), and it omits every channel beyond this one market. Provided those omitted spillovers are positive, it is a lower bound on the economy-wide value of the same R&D dollar. Consistent with that reading, the gross figure sits comfortably at the lower end of their band. The second comparison is the ratio of social to private returns. [Bloom et al. \(2013\)](#) use a firm panel to separate knowledge spillovers to technological neighbors from business stealing among product-market rivals, converting the estimates into private and social returns: their gross social returns run roughly 1.5 times private under IV and 2.7 under OLS. Their wedge prices the cross-firm knowledge-spillover channel, while my model focuses on inappropriable within-market surplus.

Overall, by an alternative route, my estimates confirm the literature's prior: the return to innovation is high, and the gap between private and social returns is wide. A gap this wide naturally puts a question on the table: can public support convert any of it into surplus? I take up this question in [Section 6](#) by simulating alternative support instruments matched on a common ex ante budget. The comparison shows how the market's product-line structure and the firms' dynamic interaction generate equilibrium effects beyond the public dollar's direct impact.

It is important to highlight that while a wedge is the canonical economic rationale for public support, it is seldom in reality the only motivation for large-scale intervention, and often not the main one. Governments also press industrial policy on grounds of geopolitical tension, supply-chain resilience, and externalities. Technological leadership itself is sometimes the policy target. My analysis remains positive and informs policy choices motivated by these concerns: it measures what the public dollar buys, an input to those choices rather than a verdict on them. I discuss the framework's broader policy implications, beyond my numerical results, in [Section 6.4](#).

⁴¹An early example is [Trajtenberg \(1989\)](#), who estimates the demand of US hospitals and clinics for computed tomography scanners.

6 Effects of Public Support

I now turn to public support for innovation. Governments have recently begun committing large packages of public money to such policies. The CHIPS and Science Act of 2022, for example, provides \$52 billion for the semiconductor industry, including \$39 billion in manufacturing incentives and \$13 billion for R&D and workforce development.⁴² To study what such an intervention buys, I simulate three stylized support instruments in the estimated GPU market, each committing \$5 billion over 2011–2020, and compare market outcomes. The three instruments are chosen to offer a push-versus-pull comparison: on the push side, an aggressive constant-rate R&D tax credit that lowers the effective cost of research, and on the pull side, two alternatives that subsidize sales of a targeted part of the product line. One pays a rebate per unit sold on the premium tier of both firms, and the other pays a rebate per unit sold on the non-premium tiers (the low-end and midrange tiers of both firms). The premium rebate is the model analog of demand pull concentrated on cutting-edge products, such as tax credits limited to the most efficient heat pumps (as in the Inflation Reduction Act); the non-premium rebate is the analog of adoption subsidies that cover products priced below a cutoff, such as electric-vehicle purchase credits.

I find that the non-premium rebate delivers the most within-market surplus per public dollar and the R&D credit induces the most research. By contrast, the premium rebate, the instrument paid on frontier products, lowers R&D and slows frontier advancement. Section 6.4 discusses the policy implications of my results.

6.1 Policy Design and Reporting

Each instrument is evaluated by re-solving the estimated environment under it and comparing outcomes to the estimated equilibrium. Each instrument is permanent; I report spending and outcomes over its first 10 years. The design holds the ex ante budget fixed and varies only its form.

Formally, the R&D credit is a reduction in the R&D-cost scale χ : under a credit at rate τ , a booked R&D dollar costs the firm $\chi - \tau$ rather than χ , and the government pays the difference. The credit's outlay in a quarter is τ times the firms' booked R&D spending that quarter. The estimated scale already embeds the public support in force over the sample period, principally

⁴²The act was signed in August 2022, under the Biden administration, and criticized by President Trump as wasteful: in his March 2025 address to Congress, President Trump called it “a horrible, horrible thing” (“We give hundreds of billions of dollars and it doesn’t mean a thing”) and urged that it be scrapped. In Intel’s case, the administration pursued a different instrument: in August 2025, Intel announced an agreement exchanging \$8.9 billion in CHIPS funding for a 9.9% government equity stake. Public equity in private firms is a subject of its own, and I do not study that instrument here.

the economy-wide federal R&E credit, so the R&D credit stacks on the existing credit rather than landing on a blank market. A rebate is a per-unit payment b to the firm on every covered unit it sells, so it lowers the covered products' effective marginal cost by b . Its outlay in a quarter is b times covered units sold. Each instrument's rate is a constant, chosen so that planned outlay, evaluated at estimated-equilibrium behavior, equals the same present-value budget B :

$$\tau \sum_{h=0}^{39} \beta^h \mathbb{E}[I_{t+h}] = b \sum_{h=0}^{39} \beta^h \mathbb{E}[Q_{t+h}^{\text{cov}}] = B,$$

where I_{t+h} is the two firms' combined R&D spending and Q_{t+h}^{cov} the covered units sold, both along the estimated-equilibrium path (expectations over simulated paths), and B is the present value of the common budget, spread evenly across the 40 quarters. The R&D credit leaves the stage game unchanged and runs on the surrogates of Section 4.3.2 as estimated; each rebate changes the stage game itself, so prices, shares, qualities, and profits under it come from surrogates refit with the rebate applied to its covered tiers (Appendix C.5).

At the \$5 billion budget, the R&D credit is a substantially more aggressive version of the federal credit the firms already receive: about an 11% reduction in the estimated R&D-cost scale, equivalently about \$0.58 per dollar of the firms' accounting R&D spending. The premium rebate comes to about \$81 per premium unit, and the non-premium rebate to about \$10.30 per covered unit. The applied rate triggers an equilibrium response in price, quality, and R&D. Realized outlay, therefore, is an equilibrium outcome rather than a constant. The credit raises R&D spending and a rebate raises covered sales, so realized spending drifts from plan. I match the budget *ex ante* (the plan comes to about \$4.1 billion in present value) because this keeps the instrument a primitive of the experiment. It is also simpler than an *ex post* match, which requires solving a computationally intensive fixed-point loop and is conceptually less transparent. Returns are computed on the realized spending.

I report three per-dollar objects: within-market surplus per public dollar, additionality, and frontier advancement per public dollar. The first two are

$$\text{surplus per public dollar} = \frac{\sum_{h=0}^{39} \beta^h \Delta W_{t+h}}{G}, \quad \text{additionality} = \frac{\sum_{h=0}^{39} \beta^h \Delta I_{t+h}}{G},$$

where ΔW_{t+h} is the change in the within-market surplus W of Section 5 in quarter $t + h$ relative to the estimated equilibrium, averaged over simulated paths, ΔI_{t+h} the corresponding change in

the two firms' combined R&D spending, and G the instrument's realized discounted outlay: the quarterly outlays defined above, discounted and summed over the horizon.⁴³ Frontier advancement per public dollar is the percent change in the industry frontier at the end of the horizon, per billion dollars of realized outlay. The public outlay cancels out of the surplus measure W : it adds to the firms' profit Π as a public transfer and is deducted as the government's cost, so it enters the per-dollar objects only as the denominator G . A public dollar is treated as a pure transfer, with a marginal cost of public funds of one; hence zero is the break-even point of surplus per public dollar: a positive value is a net gain to consumers, firms, and the government together.

6.2 Welfare Analysis

Across the three instruments, the non-premium rebate delivers the most within-market surplus per public dollar. Table 4 reports a return of 1.09, compared with 0.51 for the R&D credit and 0.23 for the premium rebate. Consumers gain under every instrument, and so do the firms once public transfers are counted.

There is a misalignment of policy ranking under the welfare and consumer surplus criteria that is worth highlighting. Specifically, the premium rebate delivers more consumer surplus per public dollar than the R&D credit, yet less welfare. The difference lies on the firms' side of the ledger: operating profit (variable profit net of public transfers) rises under the credit but falls under the premium rebate. Under that rebate, the firms receive \$8.8 billion in public transfers, but their transfer-inclusive profit rises by only \$4.4 billion: their operating profit falls by about as much, offsetting most of the \$6.0 billion consumer gain. Under the non-premium rebate, operating profit barely changes, so the consumer gain makes up almost all of the welfare gain.

The drop in operating profit under the premium rebate is largely due to a price cut by both firms triggered by the rebate. Table 5 reports the tier-level responses behind these totals: prices, markups, qualities, and shares. Under the premium rebate (Panel B), the premium's consumer price falls by an average of \$126.3, more than the roughly \$81 rebate, and its markup falls \$36.5 even with the rebate included.

There are two reasons behind this large price drop. The first is familiar: the rebate lowers the premium's effective marginal cost, so the firms cut its price and sell more. The second is that the markup itself falls, the net result of two competing forces. The first force is a change in the

⁴³Direct expenditure is the whole of the denominator: a policy-induced change in the cost of support already embedded in the estimated equilibrium is an indirect fiscal footprint, not part of the denominator G .

Table 4. Welfare per public dollar

Instrument	Per public dollar		Discounted totals (\$B)				
	Welfare	CS	ΔCS	$\Delta \Pi$	ΔI	G	ΔW
R&D credit	+0.51	+0.40	+1.8	+5.6	+0.6	4.5	+2.3
Premium rebate	+0.23	+0.68	+6.0	+4.4	−0.5	8.8	+2.0
Non-premium rebate	+1.09	+1.05	+4.8	+4.9	+0.1	4.6	+5.0

Notes: Welfare and CS per public dollar are the changes in the within-market surplus W and in consumer surplus per dollar of realized public outlay. Realized outlay G is the public outlay actually spent over the 10-year horizon; the common planned budget is about \$4.1 billion in present value. Each instrument's rate is in Section 6.1. All dollar amounts are discounted 40-quarter totals, expected outcomes averaged over simulated equilibrium paths; the Δ columns are changes relative to the estimated equilibrium, measured within the market, with downstream spillovers excluded. $\Delta \Pi$ is the change in the firms' operating profit plus the public transfers they receive, and ΔI is the change in their R&D spending. The relation $\Delta W = \Delta CS + \Delta \Pi - \Delta I - G$ is an accounting identity rather than a decomposition into independent channels. Entries are independently rounded. The transfers count in both $\Delta \Pi$ and G and cancel: a transfer that changed no behavior would leave ΔW at zero. A public dollar is treated as a pure transfer; at a marginal cost of public funds $m > 1$, the welfare-per-dollar column falls by $m - 1$ for every instrument, which leaves its ranking unchanged.

premium's buyer composition. Without the rebate, the product line sorts the least price-sensitive buyers toward the premium and the rest toward cheaper tiers or out of the market. The rebate draws buyers into the premium who are more price-sensitive than the tier's existing buyers,⁴⁴ and because a product's markup falls as its buyers become more price-sensitive, this change in composition pulls down the premium's equilibrium markup. The second force works the other way: the tier's existing buyers each become more likely to choose the premium and respond even less to its price, which tends to raise the markup. Because the premium tier's share is small, the change in buyer composition outweighs the existing buyers' response, and the premium's markup falls.

The premium price cut also draws buyers from the lower tiers and from the outside option: both lower tiers lose share, while total discrete-GPU adoption rises by about 3.4 percentage points. Consumers gain from lower premium prices and wider adoption. The firms sell more premium units but earn less on each, and they lose lower-tier volume. At the uncovered midrange, the price falls \$15.3 and the markup \$20.8, so the firms lose margin there as well; at the low-end, the price rises about as much as marginal cost, leaving the markup nearly unchanged. Premium quality, which equals each firm's own frontier, also falls on average, reflecting the cut in R&D examined in

⁴⁴This sorting intuition is akin to self-selection in the nonlinear-pricing model of [Mussa and Rosen \(1978\)](#). In the BLP demand of Section 3.1, the sorting is probabilistic rather than exact because of the type-I extreme-value taste shocks. When I re-solve prices at the observed menus with and without the premium rebate, premium buyers' mean price sensitivity $|\alpha_i|$ rises from 9.0 to 11.0 per \$1,000, and the buyers the rebate adds average 12.6 (Figure E.1).

Table 5. Tier-level equilibrium response under public support

Tier	Δ price (\$)	Δ markup (\$)	Δ quality (%)	Δ share (pp)
<i>Panel A. R&D credit</i>				
Low-end	+1.5	+0.8	+2.1	+0.41
Midrange	+1.9	+1.0	+2.1	+0.44
Premium	+19.8	+6.4	+3.1	−0.03
<i>Panel B. Premium rebate</i>				
Low-end	+14.0	+0.9	+3.9	−1.64
Midrange	−15.3	−20.8	+0.8	−0.79
Premium	−126.3	−36.5	−1.9	+5.84
<i>Panel C. Non-premium rebate</i>				
Low-end	−7.9	−8.9	+5.0	+1.92
Midrange	−45.7	−40.0	+2.2	+2.43
Premium	−8.5	−11.4	+0.6	−0.48

Notes: Entries are 40-quarter averages over simulated paths, relative to the estimated equilibrium. The price, markup, and quality columns report simple averages across both firms' products within each tier. The share column sums both firms' share changes within each tier; shares are relative to the whole market, including the outside option. Under the two rebates, the price column is the price consumers pay. The markup is the price the firm receives minus predicted marginal cost; on a covered tier, the firm receives the consumer price plus the per-unit rebate.

Section 6.3.

The non-premium rebate also lowers prices and markups (Panel C). This instrument, however, returns the most surplus per public dollar. The reason is the expanded adoption: the covered tiers gain buyers chiefly from the outside option. Total discrete-GPU share rises about 3.9 percentage points, while the premium loses only 0.48 points. Both covered tiers offer higher quality at lower prices.

Unlike the two rebates, the R&D credit reaches the product market only through innovation (Panel A). The firms raise R&D, and the resulting frontier advances improve premium quality and lower the cost of upgrading the interior tiers, so quality rises throughout the menu. Below the premium, prices change little and sales expand. At the premium, the price rises by about \$20 per unit but the markup by only about \$6, so the price rise mostly reflects higher quality, not a wider margin. Consumers gain from better products, and operating profit rises as the firms sell the improved menu. Because the R&D credit does not change the product-market equilibrium of the stage game, its product-market effects are solely due to how it changes equilibrium state visitation in the dynamic game. Its effect can therefore be roughly thought of as accelerating technological

Table 6. Static and dynamic components of welfare per public dollar under the two rebates

Instrument	Static welfare effect	Dynamic state response	Welfare per public dollar
Premium rebate	+0.43	−0.20	+0.23
Non-premium rebate	+1.00	+0.09	+1.09

Notes: The static welfare effect holds R&D and the industry frontier along the estimated-equilibrium path, with the same initial state and random draws, replays only the rebate’s within-period pricing and quality responses, and is expressed per realized public dollar of that replay. The dynamic state response is the remainder once R&D, the frontier, and market positions respond; the two sum to the total, the within-market surplus per public dollar of Table 4. The R&D credit changes no stage-game object and has no static component, so only the two rebates appear. Conventions are those of Table 4.

progress, while the equilibrium prices and qualities at each state are unaffected.

In contrast to the R&D credit, the two rebates affect welfare through both the static product-market equilibrium and the dynamic innovation equilibrium. To separate the two components, I hold R&D and the frontier at their estimated-equilibrium levels and replay each rebate’s within-period pricing and quality responses. Table 6 splits each rebate’s welfare per public dollar into the static welfare effect of this replay and the dynamic state response. This decomposition reveals that the dynamic state response can go in the opposite direction from the static welfare effect. Both rebates raise welfare in the static replay. Under the premium rebate, the dynamic state response takes back about half of the static welfare effect; under the non-premium rebate, it adds slightly to it. Section 6.3 traces these dynamic responses to R&D and the frontier.

6.3 Effects on Innovation

Table 7 reports additionality, frontier advancement per public dollar, and the percent change in R&D at the frontier-leader and follower roles under each instrument. The R&D credit induces the most private research, with additionality of about 0.13, and advances the frontier the most. The premium rebate, which targets the frontier itself, crowds out R&D and lowers the end-of-horizon frontier by 0.32% per \$1 billion of outlay. The non-premium rebate changes research little and modestly advances the frontier.

The negative additionality of the premium rebate warrants an explanation. As shown in Section 6.2, the premium rebate raises the firms’ transfer-inclusive profit by \$4.4 billion, yet their combined R&D falls; the cut lands more heavily on the frontier leader than on the follower.

Table 7. Additionality, frontier advancement, and R&D by frontier role under public support

Instrument	Additionality	Frontier advancement % per \$1B outlay	Percent change in role R&D	
			Frontier leader	Frontier follower
R&D credit	+0.13	+0.81	+8.9	+8.0
Premium rebate	−0.05	−0.32	−9.3	−3.8
Non-premium rebate	+0.02	+0.19	+2.5	+0.8

Notes: Additionality is the discounted change in the firms' private R&D per dollar of realized public outlay. Frontier advancement is the percent change in the expected end-of-horizon industry frontier $q_t^{\max}$, the higher of the two firms' frontier qualities taken within each simulated path before averaging, per \$1 billion of realized public outlay. The role columns are the percent change in each role's discounted R&D relative to its estimated-equilibrium level, about \$3.4 billion at the leader and \$3.7 billion at the follower in present value. Frontier leader and follower are state-contingent roles, not fixed firm identities: at each path-period, the leader is the firm with the higher frontier quality on the estimated-equilibrium path, and a firm's induced R&D is attributed to its role at that path-period, so a policy-induced role switch cannot relabel a dollar. The role entries describe incidence by role, not a behavioral effect of becoming a leader or follower. Conventions and realized outlays are those of Table 4; each instrument's rate is in Section 6.1.

Higher profit and less research do not intrinsically contradict each other: R&D responds not to the level of profit but to the value gap in equation (3.10), the difference between the firm's continuation values with and without an advance.⁴⁵ A local flow-profit calculation illustrates how the premium rebate changes the incentive to innovate through the product-line structure. Holding other qualities and the rival's prices fixed, with the firm's own prices at an interior optimum, the flow-profit return to a frontier advance is

$$\frac{d\pi_{ft}}{dq_{ft}^{\max}} = M_t \left[\underbrace{s_{ft}^{\text{prem}} \left(\frac{\gamma}{\tilde{\alpha}_{ft}^{\text{prem}}} - \frac{d\text{MC}_{ft}^{\text{prem}}}{dq_{ft}^{\max}} \right)}_{\text{premium channel}} + \underbrace{\sum_{j \in \mathcal{J}_f^{\text{int}}} s_{jt} \left(-\frac{\partial \text{MC}_{jt}}{\partial q_{ft}^{\max}} \right)}_{\text{cost-spillover channel}} \right], \quad (6.1)$$

where π_{ft} is the firm's transfer-inclusive variable profit conditional on the period's shocks, and the premium's cost derivative is total because its quality is the frontier. The term $\tilde{\alpha}_{ft}^{\text{prem}}$ is a weighted average of the price sensitivities $|\alpha_i|$ of the firm's premium buyers.⁴⁶ The ratio $\gamma/\tilde{\alpha}_{ft}^{\text{prem}}$ captures the pricing value of further quality, accounting for substitution within the firm's menu. Intuitively, the

⁴⁵To see why R&D need not move with the level of profit, consider a lump-sum transfer that pays the firm, or taxes it, the same amount in every state, whatever either firm does. Such a transfer shifts the value function in equation (3.8) by a constant: the firm's values with and without an advance rise or fall together, the gap between them is untouched, and the firm faces exactly the same R&D problem as before. The claim requires that the transfer not move a binding limit on R&D spending. In the model, a firm spends at most its expected variable profit on R&D, so the claim holds whenever that ceiling does not bind.

⁴⁶Consumer i 's weight is the probability that i buys the firm's premium times the premium's margin net of the firm's expected margin from i across its menu, so the weights are endogenous to the product-market equilibrium. Appendix E.3 derives equation (6.1) and the weights.

premium channel earns the value of further quality net of its production cost on premium sales; the cost-spillover channel earns production-cost savings on lower-tier sales.⁴⁷

The premium rebate changes three parts of this calculation. First, it draws buyers into the premium tier from the lower tiers and from the outside option, expanding the volume on which better quality earns a return. Second, the buyers it attracts are more price-sensitive than the tier's existing buyers. With a common quality coefficient, these buyers also value extra quality less in dollars: a buyer with coefficient α_i would pay $\gamma/|\alpha_i|$ for an additional unit of quality. A constant rebate does not change that valuation: it brings more buyers to today's flagship without making its successor more valuable to each buyer. Because $\tilde{\alpha}_{ft}^{\text{prem}}$ averages the price sensitivity of the premium's buyers, this shift in buyers can lower $\gamma/\tilde{\alpha}_{ft}^{\text{prem}}$.⁴⁸ Third, it draws buyers away from the lower tiers (their combined share falls by 2.43 percentage points in Panel B of Table 5), reducing the volume on which frontier advances save production costs. Expanded premium volume can raise the return to an advance, while buyer sorting and lost lower-tier sales can lower it. The net effect is therefore theoretically unsigned.

The non-premium rebate expands the lower-tier sales documented in Section 6.2, and with them the volume on which the cost-spillover channel earns; the premium's small loss of volume works the other way. In equilibrium, R&D changes little and the frontier edges up.

The R&D credit acts directly on the cost side of equation (3.10): unlike the two rebates, its effect on innovation is positive at any given value gap. Its additionality is nonetheless modest (0.13), because the credit pays about \$0.58 on every booked R&D dollar, including R&D the firms would do anyway. A booked dollar, moreover, costs the firm more than a dollar in economic terms (Section 3.2.2), so the credit lowers the cost of research by only about 11% (Section 6.1).

6.4 Discussion and Policy Implications

The analysis so far has traced how each instrument works through prices, qualities, and R&D. Stepping back from those mechanics, I draw several takeaways for the design and evaluation of public support for innovation.

⁴⁷Equation (6.1) illustrates the forces at play in a given state x_t at the pricing step and partly maps out how the demand and cost primitives influence their strength. In the full equilibrium, the incentive to innovate also depends on the firms' strategic responses in prices and qualities, on the evolution of the exogenous states, and on dynamic competition, all through the value gap of equation (3.10). A closed-form expression for this incentive in terms of primitives is not available without further simplifying assumptions, and a quantitative decomposition would require an intricate counterfactual design (shutting down one channel at a time). Both exercises lie outside of the scope of this paper.

⁴⁸As in Section 6.2, I re-solve prices with and without the premium rebate at each quarter's observed menu in 2011–2020, holding qualities fixed. At the re-solved prices, $\tilde{\alpha}_{ft}^{\text{prem}}$ rises under the rebate for both firms in every quarter.

First, in product-line markets, a subsidy for the frontier product can backfire and slow innovation. The frontier product's profitability level and the incentive to improve its quality need not move together. A rebate at the top of the line expands premium sales, which on its own raises the incentive to innovate. But the product line sorts the least price-sensitive buyers to the top, so the rebate draws in buyers who value further quality less in dollars. It also drains sales from the lower tiers, where an advance saves production cost.⁴⁹ Buyer sorting and lost lower-tier sales can drag the incentive down, so which way such a subsidy moves innovation is an empirical question. Here it slows innovation: the premium rebate I simulate adds \$4.4 billion to the firms' transfer-inclusive profit, yet it crowds out private R&D and holds back the frontier. The channels behind this backfire are much less salient in pharmaceuticals: a drug is rarely sold in quality tiers, and what it can treat is largely fixed at approval, with improvements arriving as new drugs. Much innovation, however, happens in markets with product lines and continual upgrades, such as consumer electronics, in which these channels can attenuate or even reverse a policy's intended effect.

Read literally, the principle of targeting, that an instrument should aim directly at its objective, seems to fail here: the premium rebate aims at the frontier itself, yet slows its advance. Under another reading, the principle still holds for my results: policy should aim as close as possible to the *decision margin*, not to the technological object. Among the three instruments, the R&D credit is the most effective at encouraging innovation: it induces the most private research and advances the frontier the most. It is also more robust: unlike the two rebates, which reach R&D only through the product-market equilibrium, the credit lowers the cost of research directly. Its direct effect is therefore signed: at any given value gap, a lower cost of research raises R&D.

Second, additionality is an incomplete yardstick for choosing among instruments. It measures the R&D a public dollar induces and leaves out the responses that largely decide an instrument's return: prices, product choices, and adoption. In every row of Table 4, induced R&D is the smallest dollar total, never more than \$0.6 billion, against consumer gains of up to \$6.0 billion. Additionality ranks the R&D credit first, yet the non-premium rebate, which it reads as adding essentially no R&D, returns the most surplus per public dollar, 1.09 against the credit's 0.51. Additionality is near zero for both rebates, -0.05 and 0.02, while their returns sharply diverge, 0.23 against 1.09. For an instrument aimed at R&D, the R&D response is a natural measure. But if the ultimate policy

⁴⁹This within-firm cost spillover is not unique to chips. An electric-vehicle maker that raises battery energy density to extend the range of its high-end models, for example, can carry the same cell technology into its mass-market models, in which a given range then takes fewer cells. Likewise, a developer of large language models can distill its frontier model into smaller ones that serve its cheaper tiers at a fraction of the cost.

objective is welfare instead of R&D per se, additionality can be a poor proxy for it, as it is here.

Third, a large social return to R&D does not mean that public support will return on a similar scale. The social return is an average over the whole R&D program, whereas a public dollar works at the margin, where it induces little additional R&D (Section 6.3). In the desktop GPU market over 2011–2020, a dollar of the industry’s R&D returns \$8.95 of within-market surplus and \$3.00 of net private return to the firms, a wedge of \$5.95 (Section 5). A public dollar spent through the R&D credit returns \$0.51. The wedge may seem a natural guide to what support can achieve, and there is now near consensus among economists that it is large. While the wedge supplies the rationale for support, it is not a measure of what a public dollar will buy. Innovation policy can go to prominent markets in which innovation has already succeeded. Those may be exactly markets like the one I study, in which high private returns already sustain heavy R&D spending: the average return to R&D is high, but a public dollar returns far less.

Finally, my model provides a framework for positive economic analysis of policies whose motivation is not mainly economic. Large industrial policies today are often motivated by national security, geopolitical tension, supply-chain resilience, or technological leadership in its own right, and the value of achieving those objectives is hard to quantify in welfare terms. What the framework can measure is instead the price of pursuing them. Whatever the objective, the instrument works through the product market in the same way, and the analysis proceeds unchanged; what changes is the reading of its output, from a verdict on the policy to the cost of the objective. For example, requiring that strategic goods such as advanced chips be produced domestically can raise their marginal cost; the increase enters the model as, in effect, a tax on production. The surplus consumers and firms surrender, and any innovation forgone over the longer run, are then not failures of the policy but its price. The framework can therefore be used to ask which instrument achieves a given non-economic goal at the least economic cost, or with the least disruption to the market.

7 Conclusion

This paper develops a framework for analyzing the private incentive to innovate and the effects of public support: a dynamic oligopoly model of product-line markets with endogenous pricing, quality positioning, and R&D. I apply it to the desktop GPU market, quantify the surplus-based private and social returns to R&D, and simulate three stylized support instruments on a common ex ante planned budget. The returns are large, and most of the value a dollar of R&D creates is

consumer surplus the firms cannot appropriate. Yet the policy effects are mixed and modest at best: a public dollar routed through any of the three instruments buys far less surplus than a dollar the firms spend on R&D themselves. A subsidy for frontier products can even backfire on innovation. The broader lesson is that pricing, positioning, and R&D move together in product-line markets: an instrument aimed at any one margin lands on all three, and the R&D it induces is an input response, not a welfare verdict.

Three future directions follow. The first is to take the framework, extended beyond two firms and three tiers, to other innovation-driven markets. A recent example is tiered access to large language models. The payoff extends beyond any single market. Growth accounting prices the economy's entire innovation effort at one average return. Market-by-market estimates would recover the distribution beneath that average, each market carrying its own social-private split and policy counterfactual.

The second direction is to measure what the within-market boundary leaves out: the chips sold to render games became the hardware that trains modern AI. Quantifying the downstream surplus created by advances in the GPU frontier would show how far the economy-wide value of an R&D dollar exceeds the within-market measure.

The third direction concerns the yardstick itself. Additionally, the R&D response a public dollar induces, is the most readily available indicator: R&D spending appears in financial reports and tax records. But the equilibrium response that decides what a public dollar ultimately buys took the full apparatus of this paper to see, machinery few agencies can build. The practical question is how to build better yardsticks where detailed data are limited: when the input response alone is a reliable guide, and what else agencies should collect when it is not. Tier-level prices, qualities, and market shares are observable for each program separately, and the product-market response shows up in them. An estimated framework of this kind offers a testbed: candidate indicators built from such data can be scored against its welfare and frontier outcomes before any is trusted in the field.

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A The Desktop GPU Market, 2005–2020

Table A.1 summarizes the product line and the market aggregates. Panel A gives the tier ladder: low-end cards account for 62.5% of units at a mean price of \$123, midrange cards for 30% at \$297, and premium cards for 7.6% at \$722, with the two lower tiers delivering about 12% and 54% of premium benchmark performance.

Table A.1. The GPU market: product line and market aggregates, 2005Q1–2020Q4

<i>Panel A. Product line by tier</i>			
Tier	Mean price (\$)	Relative performance	Unit share (%)
Low-end	123	11.5	62.5
Midrange	297	53.5	30.0
Premium	722	100.0	7.6
<i>Panel B. Market aggregates</i>			
Aggregate	Value	Basis	
Market revenue	12.9	\$B per year, average 2005–2020	
Discrete GPUs sold	14.6	million per quarter, average 2005–2020	
Attach rate	46.2	% of desktop CPU sales, pooled sample	
Combined GPU-segment R&D	1.2	\$B per year, average 2005–2020	
R&D-to-revenue ratio	0.09	pooled sample	
Frontier benchmark growth	98.7	multiple, 2005–2020	
Premium-tier price growth	1.9	multiple, 2005–2020	

Notes: Panel A reports means over the cleaned panel, weighted by units sold. Relative performance is normalized to the premium tier at 100. In Panel B, GPU-segment R&D is firm-reported spending allocated to the segment in proportion to segment revenue (Section 2).

Sources: Jon Peddie Research for discrete-GPU units and desktop CPU shipments; firm filings and Compustat for financials; PassMark for benchmark performance; launch and retail-price records for price; and the Bureau of Labor Statistics for the semiconductor components PPI that enters the price instruments (Section 4.1).

Figure A.1 plots the premium-tier price and the observed quality frontier year by year. The frontier in panel (b), the higher of the two firms’ premium-tier quality indices in each year, climbs as a staircase: slower years broken by generational jumps. The two largest proportional jumps, in 2008 and 2010, more than double the frontier within a year. The premium price in panel (a) moves far less, holding between roughly \$400 and \$470 from 2006 through 2013 and climbing only late in the sample. Midway through the sample, these gaming cards found a second constituency: Nvidia released CUDA, its platform for general-purpose GPU computing, in 2007, and the network behind the deep-learning breakthrough of 2012 was trained on two consumer gaming cards (Krizhevsky et al. 2012). Before the large-language-model boom of 2023–2024, however, the GPU market remained largely consumer-facing. It has since turned business-facing: large AI developers buy data-center accelerators for tens of thousands of dollars apiece, in batches of tens of thousands.

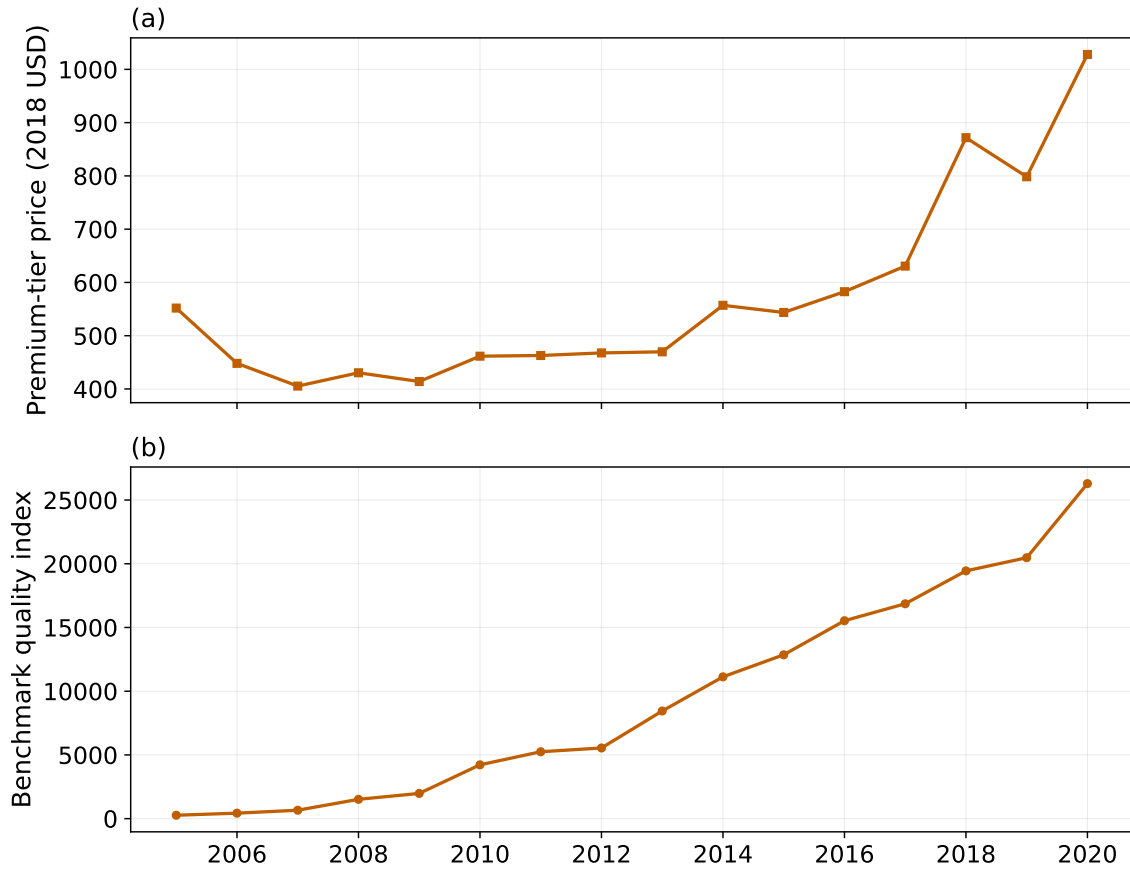


Figure A.1. Premium-tier prices and observed frontier quality

Notes: Panel (a) plots the premium-tier price, a simple mean of the two firms' premium-tier prices across the quarters within each year. Panel (b) plots the observed frontier, the annual maximum of the two firms' premium-tier quality indices, the same series the model reads as the frontier (Section 3). The 2005–2020 endpoint ratios of the two series equal the growth multiples in Table A.1, Panel B.

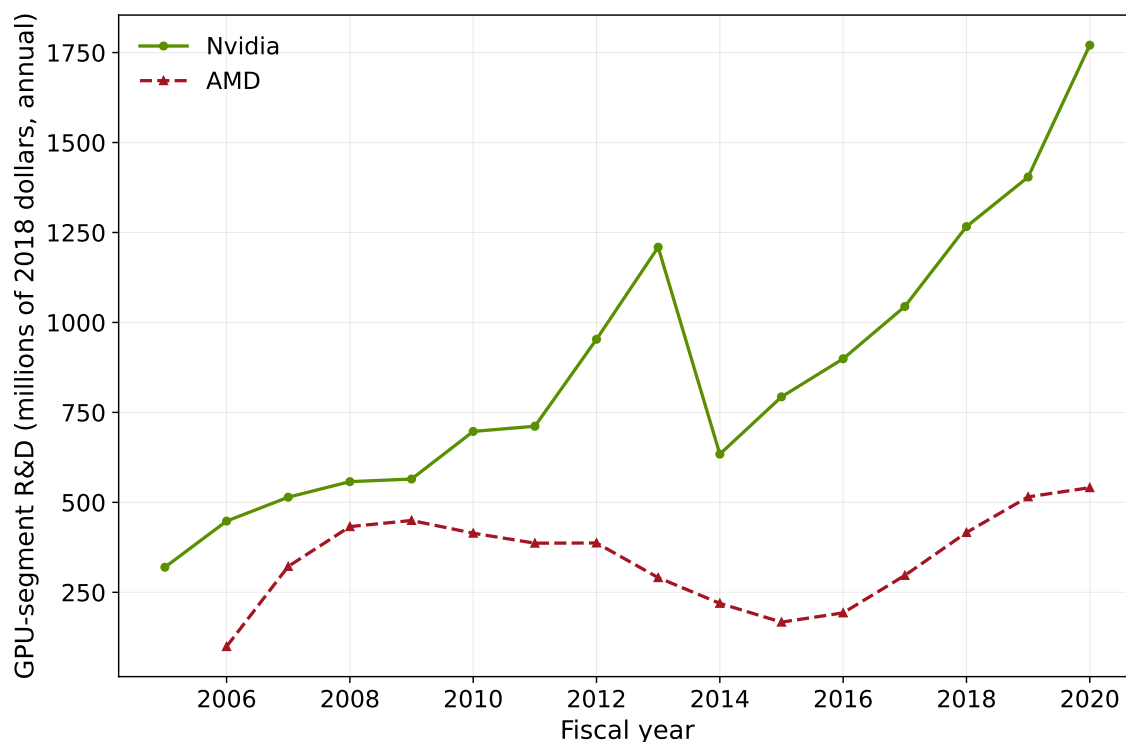


Figure A.2. Allocated GPU-segment R&D

Notes: Each series is the firm's reported R&D allocated to the GPU segment by the segment's revenue share (Section 2), summed within fiscal year, in millions of dollars; Nvidia's fiscal years are labeled by the calendar year in which they begin. Nvidia's segment definitions change several times over the sample. From fiscal 2014 onward, I use Nvidia's Gaming platform, which covers consumer products only, instead of the broader GPU segment. The rise in Nvidia's series through fiscal 2013 and the fall in fiscal 2014 are therefore partly definitional. AMD's series begins in fiscal 2006 with the acquisition of ATI, and its first year covers only the partial post-acquisition period. In estimation, the R&D-cost scale is matched to mean R&D intensity pooled across firms and quarters, rather than to the series' year-to-year movements.

The R&D behind the race appears in Figure A.2, firm by firm. Nvidia's GPU-segment R&D grows from roughly \$0.32 billion in fiscal 2005 to \$1.8 billion in fiscal 2020. AMD's runs lower throughout: it falls to roughly \$0.2 billion in fiscal 2015 and recovers to \$0.54 billion by fiscal 2020. The two series are the accounting inputs behind the R&D moment of Section 4.

Adoption and market size complete the picture. The attach rate, discrete-GPU units as a share of desktop CPU units in each quarter, averages 46.2% over the sample (Table A.1, Panel B). The denominator itself shrinks: desktop CPU shipments fall from about 37 million units a quarter in 2005 to between roughly 21 and 27 million late in the sample and level off, the maturing-market pattern that Section 4 models as mean reversion toward a long-run level.

The attach rate's largest departure from its recent range is the cryptocurrency-mining episode, the event behind the outside-option exclusion in Section 4. Over 2017Q3–2018Q2 miners bought

gaming cards. The attach rate, which had run between roughly 30% and 50% in the three preceding years, jumped to 64% in 2017Q3 and peaked at 73% in 2018Q1; the surge concentrated up the ladder, as midrange units ran roughly 70% above their early-2017 pace, premium units roughly doubled, and low-end units fell about 10%. Within the episode, prices peaked in the same quarter: in 2018Q1 Nvidia’s midrange and premium prices stood 68% and 51% above their early-2017 averages, and AMD’s more than doubled; the bump is visible even in the annual premium series of Figure A.1. The boom then unwound. In the demand system the surge registers as an outside option that is temporarily much worse: the recovered outside-option series (Figure D.1) drops in exactly these four quarters, and the trend fit excludes them.

B Details for Demand and Cost Estimation

This appendix collects the detail behind the demand and marginal-cost estimates of Section 4. Appendix B.1 reports the estimated first stage for the price variable; Appendix B.2 reports the full elasticity matrix; Appendix B.3 reports the by-firm split of Table 1’s markup panel; and Appendix B.4 constructs the implied marginal fixed cost of quality that motivates the post-2010 window of Section 4.2.

B.1 The Demand First Stage

Table B.1 reports the estimated first stage: the coefficients on the five excluded instruments with their standard errors, the included regressors, and the strength diagnostics. All five instruments enter positively, and the semiconductor components PPI loads about two and a half times as strongly on the premium tier as on the midrange tier, consistent with the die-size and yield interpretation of the tiers’ differential exposure to semiconductor input costs (Section 4.1).

B.2 Elasticities

Table B.2 reports the average elasticity matrix. The matrix is not symmetric, because the quantity response to a price change depends on the share and price of the product whose price moves. In the Nvidia midrange column, every other product’s demand rises by between 0.51% and 0.72% when that price rises by one percent; in the AMD premium column, no product’s demand rises by more than 0.20%.

Table B.1. Demand first-stage estimates

	Estimate (SE)
<i>Excluded instruments</i>	
Semiconductor PPI $\times$ premium tier	1.63 (0.22)
Semiconductor PPI $\times$ midrange tier	0.63 (0.12)
Differentiation instrument, own-firm products	6.96 (2.43)
Differentiation instrument, rival products	6.73 (1.24)
Square root of own quality	1.43 (0.46)
<i>Included regressors</i>	
Quality	Yes
Firm fixed effects	Yes
Quarter fixed effects	Yes
<i>Diagnostics</i>	
Joint excluded-instrument F statistic	48.67
Partial R^2 of excluded instruments	0.44
Observations	384

Notes: The table reports the price first stage of the maintained demand specification: a least-squares regression of price, in thousands of dollars, on the five excluded instruments and the included regressors shown. Standard errors, in parentheses, are heteroskedasticity-robust and unclustered. The joint F statistic tests the five excluded instruments; the partial R^2 measures the excluded set's explanatory contribution after the included regressors. The sample is that of the demand estimation of Section 4.1: six products over 2005Q1–2020Q4. The PPI is the Bureau of Labor Statistics semiconductor components index, lagged one quarter and demeaned over the sample.

Table B.2. Mean own- and cross-price elasticities

	AMD			Nvidia		
	Low-end	Midrange	Premium	Low-end	Midrange	Premium
AMD low-end	−1.87	0.20	0.06	0.38	0.51	0.12
AMD midrange	0.20	−3.24	0.11	0.36	0.66	0.24
AMD premium	0.14	0.20	−4.65	0.26	0.72	0.62
Nvidia low-end	0.22	0.21	0.07	−2.01	0.56	0.15
Nvidia midrange	0.19	0.23	0.13	0.34	−3.34	0.39
Nvidia premium	0.11	0.17	0.20	0.19	0.60	−4.18

Notes: Entries average product-market elasticities over the 64-quarter sample. An entry is the percent change in the row product's demand when the column product's price rises by one percent. Diagonal entries are own-price elasticities, whose tier means are in Table 1, Panel B, in absolute value; off-diagonal entries are cross-price elasticities, positive because the six products are substitutes.

Table B.3. Prices, marginal costs, and markups by firm and tier

Product	Price (\$)	Marginal cost (\$)	Markup (\$)	Lerner index
AMD low-end	121	50	71	0.59
AMD midrange	234	152	82	0.35
AMD premium	482	361	121	0.25
Nvidia low-end	143	53	90	0.63
Nvidia midrange	299	182	117	0.39
Nvidia premium	658	449	209	0.32

Notes: Entries are unweighted means over the product-quarter observations of the estimation sample, 2005Q1–2020Q4: the firm-level split behind the tier means of Table 1, Panel C. Price, marginal cost, and markup are in dollars; the Lerner index is the ratio of the mean markup to the mean price. Each column is rounded from its own unrounded series, so a printed markup can differ by \$1 from the difference of the printed price and marginal cost, and the printed Lerner index from the ratio of the printed markup to the printed price.

B.3 Marginal-Cost Recovery

Marginal costs are recovered by inverting the six pricing first-order conditions of equation (3.4) market by market at the estimated demand parameters. Table B.3 reports the recovered levels by firm and tier: for each of the six products, the mean price, marginal cost, dollar markup, and Lerner index, the firm-level split behind the tier means of Table 1. Nvidia carries the larger dollar markup in every tier, \$90 against \$71 at the low end, \$117 against \$82 at the midrange, and \$209 against \$121 at the premium tier, and the Lerner index orders the two firms the same way. The gap widens as the tier rises, so it is not a level shift common to the menu. The pattern coincides with Nvidia’s positive brand effect and its higher average quality within each tier.

B.4 The Implied Marginal Fixed Cost of Quality

The series plotted in Figure B.1 is the implied marginal fixed cost of quality introduced in Section 4.2: the left-hand side of the quality first-order condition, equation (3.6), evaluated at the observed menu for the four interior products, each firm’s low-end and midrange tiers. Under optimal quality choice, this marginal variable profit would equal any marginal fixed cost of quality, so the evaluated series is the marginal fixed cost the observed menu implies. The evaluation includes the induced response of all six equilibrium prices to each quality change. Expectations over the demand and marginal-cost shocks, ξ_{jt} and $\varepsilon_{jt}^{\text{MC}}$, are computed by sparse Kronrod–Patterson quadrature, with both shocks taken as mean-zero normal, independent across products, and scaled by the standard deviations of their estimated residuals.

Figure B.1 plots the four implied series over the full 2005Q1–2020Q4 sample and marks 2011Q1,

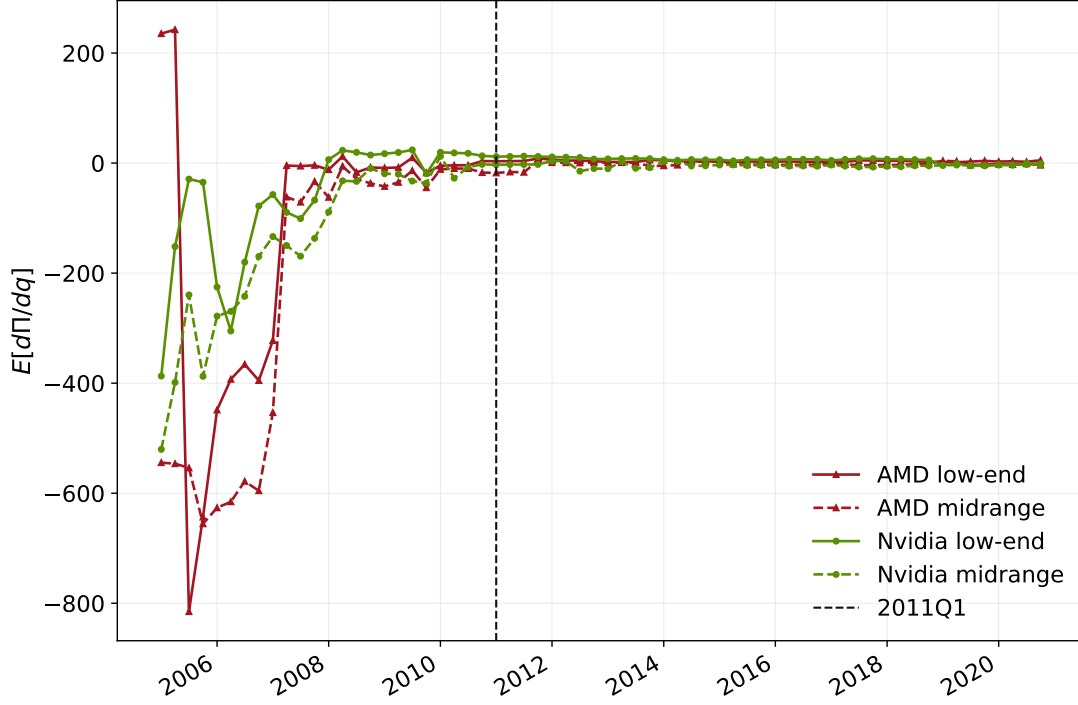


Figure B.1. Implied marginal fixed costs of quality

Notes: Each series is the derivative of the firm’s expected variable profit with respect to the product’s quality, the left-hand side of equation (3.6) evaluated at the observed menu with all six equilibrium prices adjusting; under optimal quality choice it equals the marginal fixed cost of quality. The four series cover the interior products, each firm’s low-end and midrange tiers, over 2005Q1–2020Q4. The vertical marker at 2011Q1 records the start of the maintained window, a maintained sample choice rather than an estimated break date.

the start of the maintained window. The early observations make the case for the window visible at a glance: the pre-2011 swings dwarf everything that follows, and from 2011 onward, the series settle near zero at the scale of the plot. Between 2005–2007 and 2015–2020, the mean absolute value of the series falls about 75-fold, and almost all of that decline happens before 2011. Over 2011–2020, the mean value is 0.59. Values this small are hard to interpret as more than noise in the data, so I set the marginal fixed cost of quality to zero in the dynamic estimation and the counterfactuals.

C Details for the Static-Equilibrium Approximation

The dynamic model needs the stage game’s outcomes at every state the Bellman solver and the simulated paths visit, far too many states to solve the game at each one (Section 4.3.2). Instead, I solve the stage game at a large sample of states, train constrained neural networks on those solutions, and validate the approximation in two layers: held-out accuracy (Appendix C.6) and coverage of the support the dynamic model actually visits (Appendix C.7).

C.1 The Stage Game

The stage game takes four static inputs: the two firm frontiers ($q_1^{\max}, q_2^{\max}$), the outside-option level (u_0), and the process-innovation index (z). Market size scales quantities, profits, and consumer surplus after the unit-market game (the game at unit market size) is solved. Each firm's premium quality equals its frontier, so the choice variables are the four interior qualities, each firm's low-end and midrange positions. The game is nested. Conditional on a six-product menu and a draw of the demand and marginal-cost shocks, prices solve the ownership-adjusted first-order conditions of equation (3.4). Anticipating that pricing equilibrium, qualities solve the expected-profit conditions of equation (3.6), which include the induced price response of every product in the market; fixed costs are assumed constant in quality (Section 4.2) and drop out. Interior qualities are constrained to lie between zero and the firm's frontier and to preserve the low-end, midrange, and premium ordering. The quality layer needs expected profits over the shock distribution.

C.2 Solving the Stage Game at One State

The solver first integrates over the product-level demand and marginal-cost shocks, six-vectors with independent normal components, using the same sparse Kronrod–Patterson rule as the fixed-cost inversion of Appendix B.4, so the training data and the recovered cost primitives share one shock convention. The solver then selects an equilibrium. Both the pricing and the quality games are solved from multiple starts. A candidate is admissible only if qualities are ordered and inside their frontiers and the equilibrium conditions hold up to the solver's tolerance. A converged quality-game candidate that carries a detectable profitable one-firm deviation is re-solved from that deviation until none remains, and a candidate that cannot be cleared is dropped. When several admissible equilibria remain, the selection rule keeps the one with the highest total industry variable profit, and the same rule is applied when generating training data, validating held-out states, and fitting the rebate-specific surrogates. In practice, multiplicity is negligible.

C.3 Designing the Training Sample

The base design covers two rectangular regions of states and quality menus, one bracketing the observed path and one extending beyond it, with the observed post-2010 states appended as anchors and additional draws concentrated near them. The design also appends the states on which the dynamic solvers fit the value functions, states bridging the two sampling regions, and the

boundary regions of the solvers' state box, so the static mapping is trained where the dynamic layer reads it. The same design serves the maintained specification and the sensitivity reestimations of Appendix F.

The sample is generated in two passes. The first draws quality menus and states across the domain, subject to the tier-ordering constraints, solves the pricing game at each draw, and fits the pricing network to the shock-integrated solutions; the second solves the quality game across sampled states with the fitted pricing network as the inner map and fits the quality network to the selected equilibria. The split concentrates the costly game solving where a dense sample is feasible, the pricing surface, rather than inside every quality best-response iterate.

For the rebate-specific refits of Appendix C.5, after the dynamic model is first estimated, I augment the design: the states visited by the estimated equilibrium's simulated paths are appended, so each refit trains densely where the policy paths travel. Throughout, I set the training bounds wide enough that state visitation in the dynamic estimation and the counterfactuals stays inside them, the design fact behind the no-extrapolation statement of Section 4.3.2. Appendix C.7 records the realized rates.

C.4 The Surrogate Networks and Their Economic Constraints

Two neural-network ensembles are fitted to these training sets, one for quality and one for pricing. Each member is a small feed-forward network trained to minimize squared prediction error, and each ensemble averages 20 independently initialized members with equal weight. The quality ensemble Q maps the two frontiers and the two exogenous static states into the four interior qualities:

$$\widehat{q}^{\text{int}} = Q(q_1^{\text{max}}, q_2^{\text{max}}, u_0, z). \quad (\text{C.1})$$

The pricing ensemble $\mathcal{P}$ takes the resulting six-product menu q together with the same two exogenous states and returns the shock-integrated equilibrium outcomes (expected prices, shares, variable profits, and consumer surplus):

$$\left(\widehat{p}, \widehat{s}, \widehat{\pi}, \widehat{\text{CS}}\right) = \mathcal{P}(q, u_0, z). \quad (\text{C.2})$$

Economic constraints are built into the output layers. The quality ensemble returns interior qualities as ordered sigmoid fractions of the firm frontier, so each firm's low-end and midrange qualities respect the tier ordering and stay inside the frontier by construction. The pricing ensemble

Table C.1. Held-out surrogate accuracy

Output block	Neural-network R^2	Cubic R^2	Best LASSO R^2
AMD low-end quality	0.98	0.96	0.89
AMD midrange quality	0.97	0.95	0.88
Nvidia low-end quality	0.98	0.97	0.89
Nvidia midrange quality	0.96	0.94	0.87
Interior quality, average	0.97	0.96	0.88
Prices, average	1.00	1.00	0.99
Shares, average	0.99	0.72	0.61
Variable profits, average	0.98	0.70	0.59
Consumer surplus	1.00	0.97	0.95
All pricing outputs, average	0.99	0.81	0.74

Notes: The validation split is a single 20% test split held out from training. Group averages are unweighted means across the corresponding network outputs: six products for prices, shares, and variable profits, four for the interior qualities. Both benchmarks are fitted on the training sample and scored on the same held-out split. The cubic benchmark is a degree-three polynomial fitted by least squares. Within each output block, the LASSO benchmark is the ℓ_1 -penalized polynomial whose degree (two or three) and penalty maximize held-out R^2 among the configurations compared. Values of 1.00 reflect rounding, not perfect fit.

returns the six inside shares and the outside share through a softmax over seven logits; shares are nonnegative and sum to one. Its prices pass through a positive output activation.

C.5 Rebate-Specific Refits

Each static environment gets its own fitted pricing-and-quality pair. A rebate changes the stage game itself: it enters the pricing first-order conditions on the covered tiers (Section 6). For each rebate, I re-solve the games with the rebate applied and fit rebate-specific surrogates on the same state domain, including the visited-state augmentation of Appendix C.3. The refit changes the static mapping only; the dynamic state variables, their transitions, and the R&D technology are those of the estimated environment. The R&D credit and the R&D shutdowns act on the dynamic layer and leave the stage game unchanged, so they run on the estimated equilibrium’s surrogates.

C.6 Held-Out Accuracy

Table C.1 reports held-out R^2 by output block for the estimated environment’s surrogates, beside two polynomial benchmarks. The validation sample is a single 20% split held out from training and never used to tune the networks.

C.7 Coverage of the Visited Support

Held-out accuracy certifies the fit on states like those the networks were trained on. Outside their training bounds, networks extrapolate, and held-out accuracy is uninformative there. For every simulated state, I therefore record whether the quality or the pricing mapping is evaluated outside its training bounds and whether the realized R&D actions and interior qualities sit at the bounds of the firm’s problem. The recorded rates confirm that the training bounds contain visitation: in the estimated equilibrium and every policy counterfactual, neither mapping is ever evaluated outside them, and on the estimated equilibrium’s simulated paths, state visitation stays well inside the solvers’ state box. The R&D policy is interior in nearly all simulated firm-quarters, and the realized interior qualities never sit against the feasibility bounds of the firm’s problem.

D Details for Dynamic Estimation

D.1 Exogenous-State Processes and Long-Run Closure

The three exogenous states (the outside option, the process-innovation index, and market size) evolve deterministically by the laws of motion of Section 3.2.2 (equations (3.11), (3.12), and (3.13)); this subsection records the fitted parameters and the closure bounds. The outside-option drift is recovered from the demand system’s quarter fixed effects, excluding the 2017Q3–2018Q2 cryptocurrency-mining window (Section 4.3.1). In monetary units, the fitted trend has a slope of \$24.08 per year, with $R^2 = 0.95$ over 60 quarters; converted to utility units, it gives a quarterly drift of 0.11, the rounded value at which the model is solved (Panel D of Table D.1). Figure D.1 plots the recovered series and the fitted trend, including the excluded window.

The closure bounds cap both trends. The process-innovation drift, the time component of the marginal-cost projection, extends the in-sample decline of about 0.04 per quarter until the index reaches the floor of -2.5 ; the outside option rises by its quarterly drift until it reaches the ceiling of 7.0. On the maintained laws of motion, both bounds bind about five years beyond the 2020Q4 sample end, so neither is reached on the 40-quarter reported paths. The bounds still enter through continuation values (Appendix E.1). Neither bound is a free level: each equals the fitted 2020Q4 value advanced by 20 quarters of the estimated drift, rounded to the nearest 0.5. The in-sample trend disciplines the drift, but the date at which the trend stops is not identified by the sample. Appendix F.2 treats the closure as a sensitivity dimension.

Table D.1. Dynamic parameters and calibrated processes

<i>Panel A. Estimated in the solved dynamic game</i>		
Object	Parameter	Estimate (SE)
R&D-cost scale	χ	5.13 (0.66)
<i>Panel B. Innovation arrival and jump distribution, estimated offline</i>		
Object	Parameter	Estimate (SE)
AMD success-rate parameter	a_{AMD}	-1.20 (0.43)
Nvidia success-rate parameter	a_{Nvidia}	-1.61 (0.46)
Fishing-out coefficient	φ	0.36 (3.29)
Mean log jump	m_J	-4.20 (0.17)
Variance of log jump	v_J	0.66 (0.15)
<i>Panel C. Calibrated discount factor</i>		
Object	Parameter	Value
Quarterly discount factor	β	0.99
<i>Panel D. Fitted exogenous-state processes</i>		
Object	Parameter	Value
Outside-option quarterly drift	Δ_u	0.11
Outside-option ceiling	$u_{0,\max}$	7.0
Process-innovation quarterly drift	c_2	-0.04
Process-innovation floor	$z_{\min}$	-2.5
Market-size persistence	ρ_M	0.85
Long-run market size	M_∞	28.0 million

Notes: The R&D-cost scale χ is estimated from a single moment, the mean R&D intensity of the post-2010 sample (Appendix D.3). The standard error of 0.66 is asymptotic, computed by the delta method, consistent with the objective function plotted in Figure D.2. A parametric bootstrap that redraws innovation arrivals and jumps over the sample window yields a larger standard error of 3.33, capturing the uncertainty that is intrinsic to innovation. Both constructions hold the demand, marginal-cost, frontier-transition, and surrogate objects at their point estimates; counterfactual quantities in Sections 5 and 6 are likewise reported as point estimates conditional on the same blocks. Panel B standard errors come from the offline arrival-rate likelihood and sample-moment asymptotics. The fishing-out coefficient is imprecisely estimated, a consequence of the rarity of frontier advances; Appendix F.2 reports sensitivity to the hazard specification. In Panel D the drifts, the persistence, and the long-run market size are fitted values; the ceiling and the floor are maintained closure bounds (Appendix D.1).

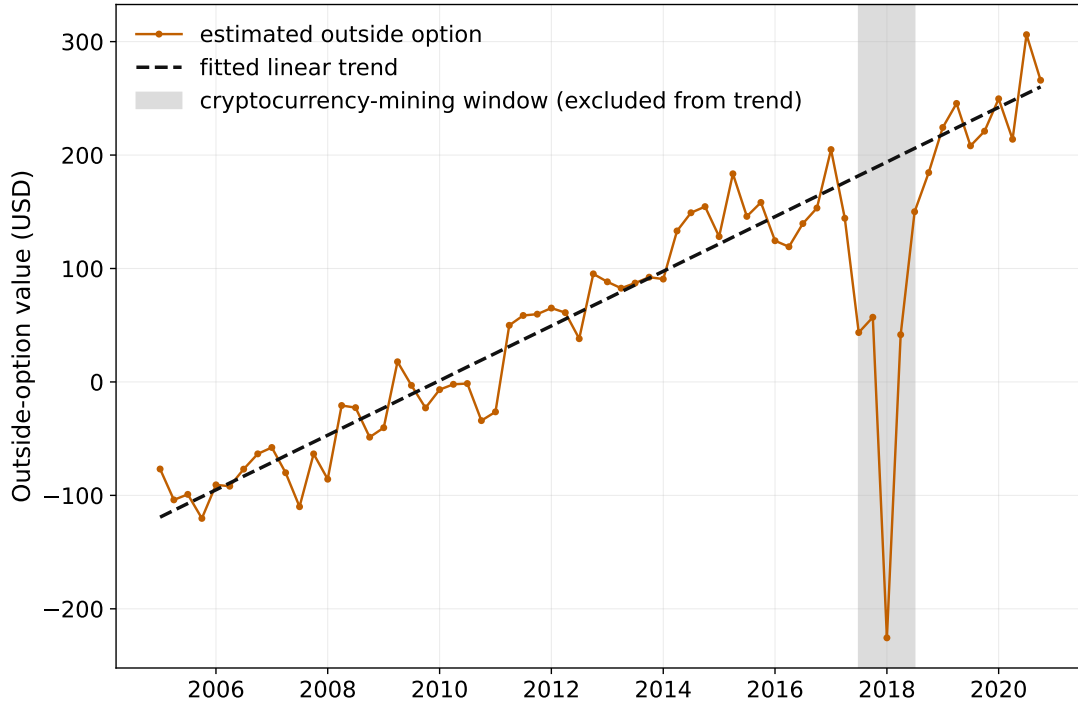


Figure D.1. Recovered outside option and fitted transition

Notes: The recovered outside-option value, plotted in dollars, is obtained from the demand system's quarter fixed effects. The linear trend is fitted after excluding the shaded 2017Q3–2018Q2 cryptocurrency-mining window.

The market-size calibration uses 2011Q1–2020Q4: a log-space AR(1) fit gives a quarterly persistence of 0.85 (standard error 0.09) and a long-run level of 28.43 million desktop units, with $R^2 = 0.72$ over 39 transitions; the model is solved at the rounded long-run level of 28.0 million. The initial market size is 36 million units, the rounded 2011–2013 mean.

D.2 Solution Method and Simulation

The solver of Section 4.3.3 enforces the Bellman equation on the observed 2011–2020 state path and on a space-filling quasi-random sample that extends beyond it. The continuation value is computed as an exact sum rather than by simulation: the solver sums over the four ways the two frontiers can move in a quarter, weighting each by its probability, and integrates over the size of an advance by quadrature. Forward simulation draws advances and jump sizes from the same distributions. Across values of the cost scale, a given path and period reuse the same success and jump draws, so differences in the simulated moment reflect the parameter rather than the draws.

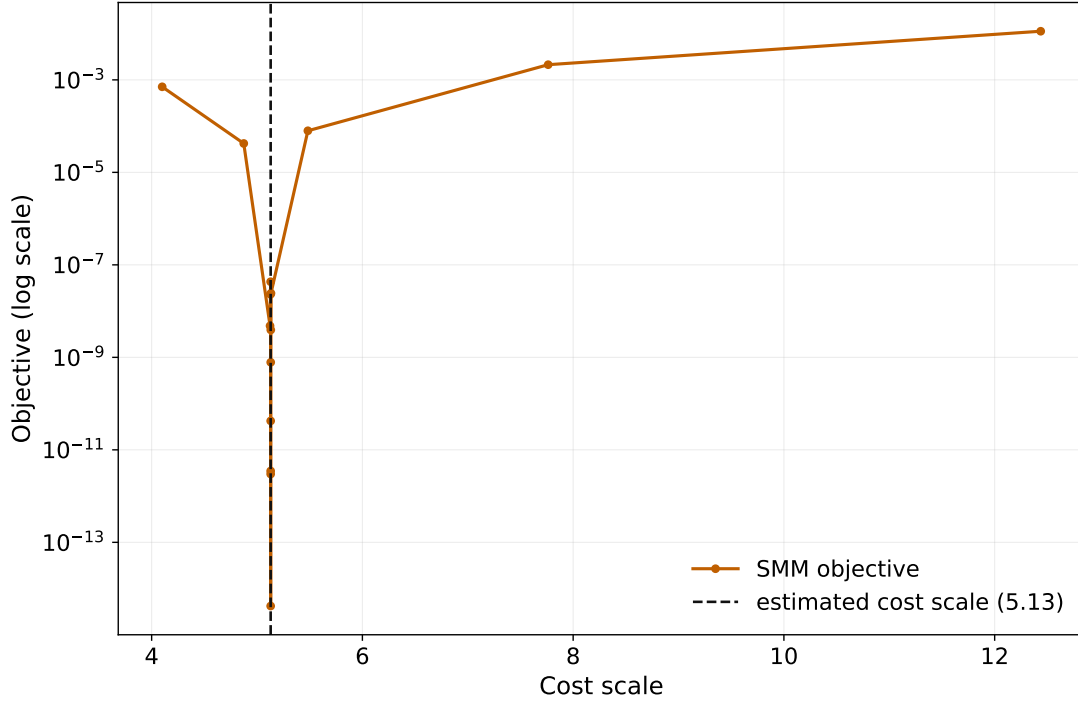


Figure D.2. The estimation objective for the dynamic R&D-cost scale

Notes: The objective is the squared gap between simulated and observed mean R&D intensity (equation (D.1)). Points come from converged equilibrium solutions and are plotted against the cost scale, connected for readability. The vertical axis is logarithmic, spreading the evaluations clustered near the minimum. The dashed vertical line marks the estimate, $\hat{\chi} = 5.13$.

D.3 The Estimation Moment and the Objective

The cost scale is estimated by matching the model-implied average R&D intensity to its observed value over 2011–2020. The objective function is the squared gap between the two:

$$\hat{m}_I = \frac{1}{2T} \sum_{f=1}^2 \sum_{t=1}^T \iota_{ft}^{\text{data}}, \quad Q(\chi) = \left[\hat{m}_I - \mathbb{E}_{\chi} \left(\frac{1}{2T} \sum_{f,t} \iota_{ft} \right) \right]^2, \quad (\text{D.1})$$

with $T = 40$ the number of quarters in the 2011–2020 window. The targeted mean intensity is 0.21, and the estimator matches it to numerical precision. The exact match is mechanical for an exactly identified estimator, so the fit evidence rests on the by-firm paths of Appendix D.4, dimensions the estimation does not use, rather than on this moment.

Figure D.2 plots the objective at the evaluated cost-scale values. The lowest value occurs at $\hat{\chi} = 5.13$.

D.4 Model Fit Beyond the Estimation Moment

The dynamic estimation matches a single moment, the pooled mean R&D intensity. Model fit is thus demonstrated by how closely the model reproduces patterns in the data it was not asked to match. I provide that check for market share, frontier quality, and R&D intensity, for both firms over time.

Figure D.3 presents the check, with firms in columns and objects in rows. In the share panels (a), the dashed line is the model-implied expected share at the observed states; in the frontier panels (b), it is the observed frontier quality; and in the R&D-intensity panels (c), it is the series that constructs the data moment, observed R&D spending over model-implied expected profit, with the exogenous states on their fitted paths. The solid line and the shaded band are the mean and the pointwise 10th–90th percentile range across 256 simulations at the point estimate. The band reflects randomness in innovation successes and jump sizes rather than parameter uncertainty. The mean series is smoother than the dashed series by construction: successes and jumps are discrete, so any single path, observed or simulated, moves in steps, and averaging across simulations smooths those steps out. Likewise, every counterfactual result I report on simulated paths is computed from averages across them, so it integrates out the randomness in the innovation process.

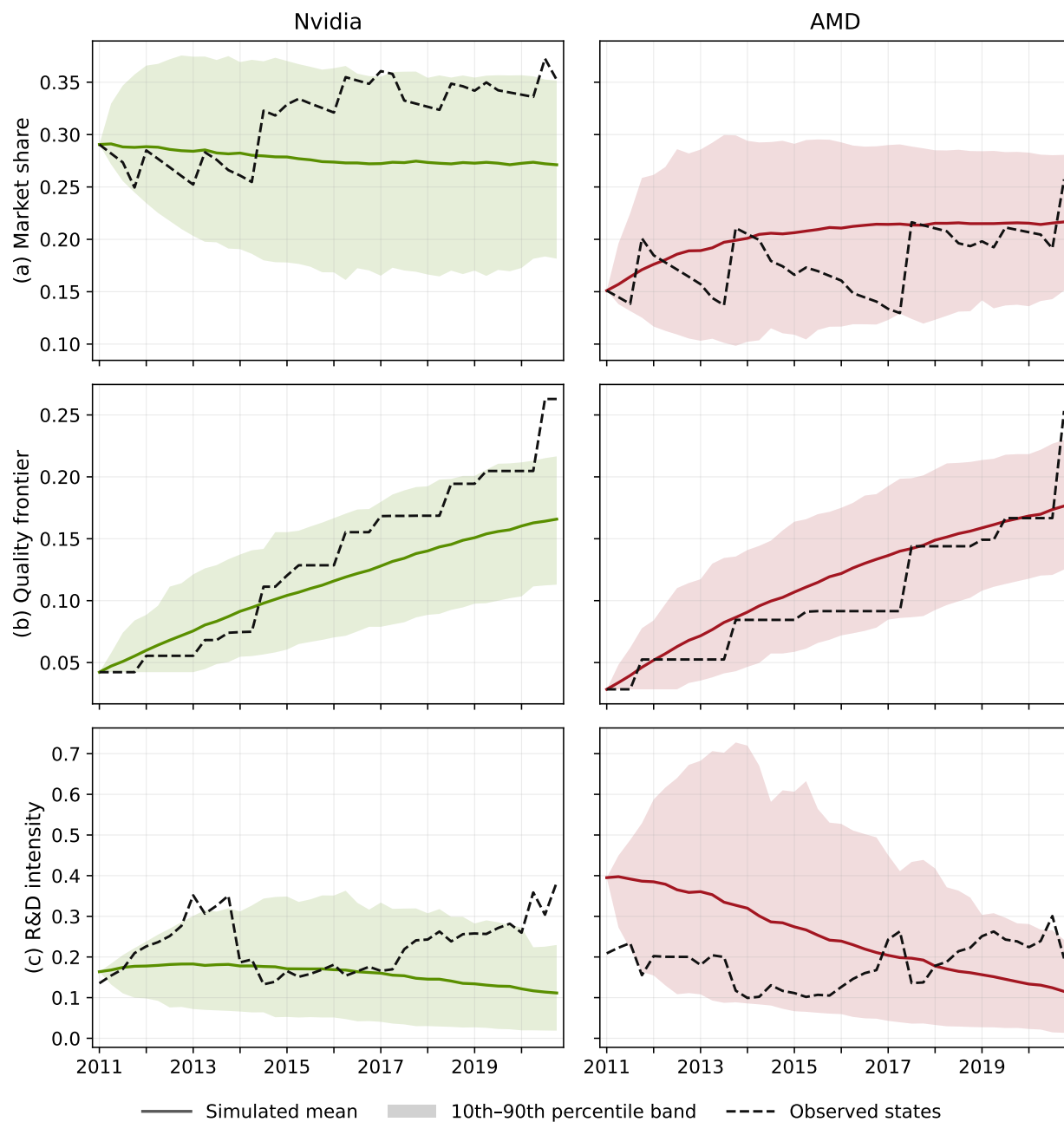


Figure D.3. Paths at the observed state path and simulated point-estimate bands

Notes: Columns are firms; rows show market share (a), the quality frontier in units of 100,000 benchmark points (b), and R&D intensity (c), with the vertical scale shared within each row. Dashed paths are constructed at the observed state path: the model-implied expected share in the share panels (a); the observed frontier quality in the frontier panels (b); and observed R&D spending over model-implied expected profit, with the exogenous states on their fitted paths, in the R&D-intensity panels (c). Solid paths show the simulated mean, with the shaded pointwise 10th–90th percentile band across 256 simulations at the point estimate; the bands reflect innovation and jump randomness rather than parameter uncertainty.

E Details for Counterfactuals and Additional Results

This appendix collects the mechanics behind the counterfactuals of Sections 5 and 6 and the evidence behind their mechanism claims. Counterfactual experiments change one economic primitive at a time and compare the resulting paths against the estimated equilibrium under a common initial state and common random numbers. The shutdown experiments ask what a firm’s own R&D buys, and the policy experiments ask what an additional public dollar buys. The estimated equilibrium throughout is the model solved at the estimated primitives, whose R&D-cost environment already embeds the historical support. Appendix E.1 records the common protocol; Appendix E.2 reports the buyer-sorting exercise behind the premium rebate’s price cut; and Appendix E.3 derives the local flow-profit return of Section 6.3.

E.1 Common Counterfactual Protocol

Each policy counterfactual is permanent in the solved game: agents form continuation values under the counterfactual law of motion rather than treating the intervention as a 10-year surprise, so the 10-year window is a reporting window, and a flow measured in the first 40 quarters can respond to continuation values beyond quarter 40. All paths start from the same 2011Q1 state: the observed frontier qualities, with the exogenous states on the fitted paths of Appendix D.1. Each regime reuses the innovation and jump-size draws of the estimated equilibrium across 256 simulated paths, so common random numbers damp simulation noise in the difference without changing the stochastic model. Beyond the horizon, every regime keeps the stationary exogenous-state closure of estimation, with no terminal scrap-value adjustment.

The fixed initial state prevents a policy from receiving credit for a favorable draw before it begins, and the shared closure prevents comparisons from mixing different beliefs about the outside option, process innovation, or market size. Appendix F.2 provides sensitivity analysis for this closure assumption.

E.2 Buyer Sorting Under the Premium Rebate

This subsection documents the buyer sorting that Section 6.2 uses to explain the premium rebate’s price cut. At each quarter’s observed menu in 2011–2020, I re-solve prices with and without the rebate, holding qualities fixed. Figure E.1 plots the resulting distribution of price sensitivity $|\alpha_i|$ among each tier’s buyers. Without the rebate, the product line sorts the least price-sensitive

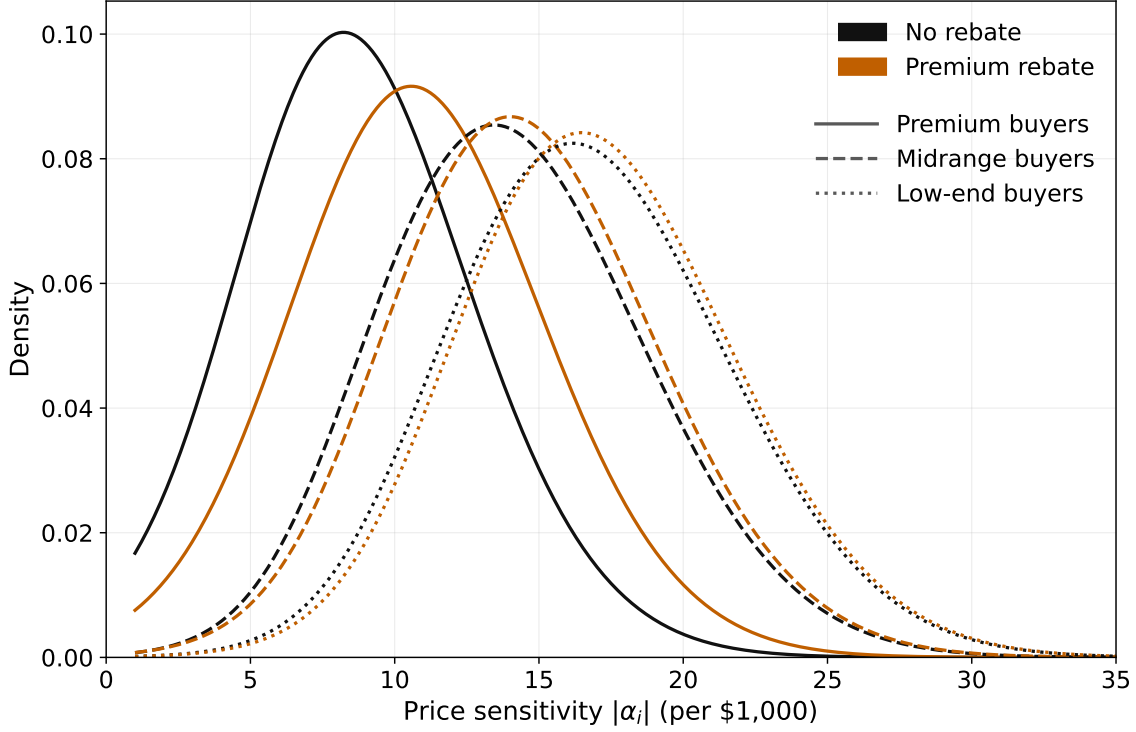


Figure E.1. Buyer sorting under the premium rebate

Notes: Each curve is the density of the price-coefficient magnitude $|\alpha_i|$, with price in thousands of dollars, among one tier's buyers: solid for the premium tier, dashed for the midrange, and dotted for the low-end. Black curves are without the rebate and orange curves with an \$80.94 rebate on each firm's premium product. At each quarter of 2011–2020, prices are re-solved in both cases at the observed qualities and frontiers, with the demand and cost shocks set to zero. Buyer composition integrates out the logit taste shocks and weights buyers by the estimated distribution of the price coefficient; it pools the two firms within each tier and pools the 40 quarters with weights proportional to market size. Each curve integrates to one within its buyer group, so its height does not measure the tier's share.

buyers toward the premium, whose curve lies furthest left. The rebate shifts that curve toward the midrange's, because the buyers it adds are more price-sensitive than existing premium buyers: 12.6 per \$1,000 on average, against 9.0. The premium buyers' mean rises to 11.0 and is higher in every quarter. Both firms' premium prices fall by more than the rebate in every quarter, so the premium's markup falls even with qualities fixed. On the simulated paths, it also falls at both firms, by 23.8% at Nvidia and 15.7% at AMD in 40-quarter averages.

E.3 Local Flow-Profit Return to a Frontier Advance

This subsection derives equation (6.1). Hold rival prices, other qualities, rebates, shocks, and exogenous states fixed, with premium quality equal to $q_{ft}^{\max}$. Here p_{jt} is the price the firm receives (the consumer pays $p_{jt} - b_d$ on a rebate-covered product), so product j 's margin is $m_{jt} = p_{jt} - MC_{jt}$,

and the firm's transfer-inclusive variable profit is $\pi_{ft} = M_t \sum_{j \in \mathcal{J}_f} m_{jt} s_{jt}$.

The envelope theorem removes the firm's own-price responses. Each response multiplies a pricing condition in equation (3.4), which is zero at an interior optimum. At given prices, the frontier moves demand through premium quality and changes production costs: the premium's quality rises with the frontier, while the interior products' qualities remain fixed. Hence

$$\frac{d\pi_{ft}}{dq_{ft}^{\max}} = M_t \left[\sum_{j \in \mathcal{J}_f} m_{jt} \frac{\partial s_{jt}}{\partial q_{ft}^{\max}} - s_{ft}^{\text{prem}} \frac{d\text{MC}_{ft}^{\text{prem}}}{dq_{ft}^{\max}} - \sum_{j \in \mathcal{J}_f^{\text{int}}} s_{jt} \frac{\partial \text{MC}_{jt}}{\partial q_{ft}^{\max}} \right], \quad (\text{E.1})$$

where a superscript prem marks firm f 's premium product. The two cost derivatives follow from equation (3.7):

$$\frac{d\text{MC}_{ft}^{\text{prem}}}{dq_{ft}^{\max}} = z_t + 2c_3 q_{ft}^{\max}, \quad -\frac{\partial \text{MC}_{jt}}{\partial q_{ft}^{\max}} = \frac{q_{jt}}{(q_{ft}^{\max})^2} \left(c_4 + 2c_5 \frac{q_{jt}}{q_{ft}^{\max}} \right), \quad j \in \mathcal{J}_f^{\text{int}}.$$

For the premium, quality rises with the frontier, so its frontier-relative cost terms remain constant. For each interior product, quality stays fixed, so only those frontier-relative terms change.

To evaluate the demand term, let s_{ijt} be the choice probability of a consumer with price coefficient α_i , the integrand of equation (3.3) evaluated at consumer prices; s_{ift}^{prem} is consumer i 's probability of buying firm f 's premium. In demand, $q_{ft}^{\max}$ and p_{ft}^{prem} enter only the premium's utility, with coefficients γ and α_i , so for every alternative j ,

$$\frac{\partial s_{ijt}}{\partial q_{ft}^{\max}} = \gamma s_{ijt} (\mathbf{1}\{j = \text{prem}\} - s_{ift}^{\text{prem}}), \quad \frac{\partial s_{ijt}}{\partial p_{ft}^{\text{prem}}} = \alpha_i s_{ijt} (\mathbf{1}\{j = \text{prem}\} - s_{ift}^{\text{prem}}).$$

Weighting by the firm's margins, summing over its products, and integrating over consumers gives

$$\sum_{j \in \mathcal{J}_f} m_{jt} \frac{\partial s_{jt}}{\partial q_{ft}^{\max}} = \gamma \int \omega_{ift} d\Phi(v_i), \quad \sum_{j \in \mathcal{J}_f} m_{jt} \frac{\partial s_{jt}}{\partial p_{ft}^{\text{prem}}} = \int \alpha_i \omega_{ift} d\Phi(v_i),$$

where $\omega_{ift} = s_{ift}^{\text{prem}} (m_{ft}^{\text{prem}} - \sum_{j \in \mathcal{J}_f} m_{jt} s_{ijt})$. This weight is the consumer's premium purchase probability times the premium margin net of the firm's expected margin from that consumer. The premium's pricing condition sets the second sum to $-s_{ft}^{\text{prem}}$, so $\int (-\alpha_i) \omega_{ift} d\Phi(v_i) = s_{ft}^{\text{prem}}$. Define

$$\tilde{\alpha}_{ft}^{\text{prem}} = \frac{\int (-\alpha_i) \omega_{ift} d\Phi(v_i)}{\int \omega_{ift} d\Phi(v_i)}.$$

Then $\int \omega_{ift} d\Phi(v_i) = s_{ft}^{\text{prem}} / \tilde{\alpha}_{ft}^{\text{prem}}$, and the demand term equals $\gamma s_{ft}^{\text{prem}} / \tilde{\alpha}_{ft}^{\text{prem}}$. Substituting the demand term into equation (E.1) gives equation (6.1).

By construction, $\tilde{\alpha}_{ft}^{\text{prem}}$ is a weighted average of the premium buyers' price sensitivities $|\alpha_i|$, with weights ω_{ift} , positive at the observed menus. Equivalently, $\gamma / \tilde{\alpha}_{ft}^{\text{prem}}$ averages the values $\gamma / |\alpha_i|$ that consumers place on a unit of quality, with weights $-\alpha_i \omega_{ift} / s_{ft}^{\text{prem}}$ that integrate to one by the premium's pricing condition. The weighted average $\tilde{\alpha}_{ft}^{\text{prem}}$ differs from the simple average of premium buyers' $|\alpha_i|$ in Section 6.2 and Figure E.1, which weights each consumer by the probability of buying a premium product alone and pools the two firms and the 40 quarters. With $\sigma_{\text{price}} = 0$, $\tilde{\alpha}_{ft}^{\text{prem}} = |\bar{\alpha}|$, which no rebate can move.

F Robustness and Sensitivity Analysis

This appendix collects the robustness checks behind the demand estimates and the R&D-cost scale of Section 4 and the returns to R&D of Section 5. Appendix F.1 varies the demand specification and carries each variant through the recovered marginal costs to the tier markups. Appendix F.2 varies the dynamic specification, reestimates the R&D-cost scale, and recomputes the returns to firm R&D under each variant. Appendix F.3 illustrates the accounting sensitivity of the estimates and counterfactual results to a common multiplier on allocated GPU R&D. Across the reestimated specifications, the mean own-price elasticity stays near 3, the R&D-cost scale keeps the economic cost of R&D well above the accounting cost, and, with the demand system held fixed across the dynamic variants, the social return to R&D remains roughly three times the net private return.

F.1 Demand-Specification Robustness

The maintained demand specification combines the excluded-instrument set of Section 4.1, quarter fixed effects, one random coefficient on price, and a time-varying market size.

Table F.1 varies the estimation sample and the market-size assumption. The crypto-quarters row removes the cryptocurrency-demand episode from the estimation sample. The market-size row replaces the time-varying market-size path with a constant. I judge the variants on the downstream objects they feed rather than on coefficient stability alone, so the table carries each through the recovered marginal costs and reports the tier Lerner indices.

The maintained specification estimates a price coefficient of -17.83 (standard error 5.30) and a quality coefficient of 60.51 (12.01), a mean own-price elasticity of 3.21, and tier Lerner indices of

Table F.1. Demand-specification robustness

Specification	Price coef.	Quality coef.	Mean own	Lerner index		
	(SE)	(SE)	elasticity $ \varepsilon_{jj} $	Low-end	Midrange	Premium
Maintained specification	-17.83 (5.30)	60.51 (12.01)	3.21	0.61	0.37	0.29
Excluding crypto quarters	-17.38 (4.68)	52.17 (8.95)	2.95	0.64	0.40	0.32
Constant market size	-18.85 (4.79)	55.97 (8.50)	2.99	0.59	0.38	0.34

Notes: Each row below the maintained specification reestimates the demand system of Table 1 after one change to that specification: the crypto-quarters row excludes the 2017Q3–2018Q2 cryptocurrency-demand quarters, and the market-size row replaces the time-varying market size with a constant. Standard errors, in parentheses, are heteroskedasticity-robust and unclustered. Mean own elasticity is the mean absolute own-price elasticity across the six products; Lerner indices are tier-mean markup over tier-mean price, from posted price and recovered marginal cost.

0.61, 0.37, and 0.29 from low-end to premium. The crypto-quarters and market-size variants stay within one standard error of the maintained coefficients and give elasticities of 2.95 and 2.99. The tier Lerner indices move by at most 0.05: across the three specifications, the low-end rate runs from 0.59 to 0.64, the midrange rate from 0.37 to 0.40, and the premium rate from 0.29 to 0.34.

F.2 Dynamic Specification and Estimation

This subsection reports the sensitivity behind the R&D-cost scale of Section 4.3.4 and the returns to R&D of Section 5: each variant changes one primitive of the dynamic specification, reestimates the scale, and repeats the full-industry return experiment. The sensitivity analysis covers the cost scale and returns; it does not re-solve the policy exercises of Section 6 under each variant. The maintained specification pairs the arrival-rate hazard with log-concave R&D exposure over the 2011Q1–2020Q4 estimation window and closes the exogenous states five years after the sample ends. Hazard and jump parameters are estimated offline on the full 2006–2020 innovation history.

Table F.2 varies one primitive per row. The curvature row lowers the exposure curvature to $\kappa = 0.25$ within the maintained hazard family. The Poisson row holds the curvature at its maintained $\kappa = 0.5$ and replaces the saturating map of equation (3.9), $h_f = r_{ft}/(1 + r_{ft})$, with the Poisson map $h_f = 1 - \exp(-r_{ft})$, the arrival rate r_{ft} entering as before. Both change the mapping from R&D and frontier position to innovation arrival. The hazard-window row instead fits the hazard and the conditional jump distribution to the post-2010 history alone and rematches χ to the unchanged R&D-intensity moment. The plateau rows defer the exogenous-state plateau to seven and ten years after the sample end, which changes continuation values through the levels at which the exogenous states settle. The return columns repeat the full-industry experiment of Section 5, in which both firms' R&D shuts down at once, at each row's reestimated scale and re-solved equilibrium.

Table F.2. Sensitivity of the R&D-cost scale and the returns to R&D

Specification	$\hat{\chi}$ (SE)	Full-industry return per R&D dollar		
		Net private	Social	Wedge
Maintained specification	5.13 (0.66)	3.00	8.95	5.95
Exposure curvature 0.25	5.23 (0.63)	2.93	8.83	5.90
Poisson arrival hazard	5.21 (0.51)	2.95	8.91	5.96
Post-2010 hazard window	3.15 (0.44)	3.53	10.69	7.16
Plateau seven years after the sample	5.46 (1.00)	2.90	8.67	5.77
Plateau ten years after the sample	5.40 (0.85)	2.81	8.50	5.69

Notes: Each row reestimates the R&D-cost scale χ from the average R&D-intensity moment after one change to the maintained specification, described in the text, and repeats the full-industry experiment of Table 3, in which both firms' R&D shuts down at once, at the row's reestimated scale; the return columns are defined as in that table's note. Parentheses report asymptotic standard errors, computed by the delta method.

The plateau rows vary an assumption rather than an estimate: the in-sample drifts of the outside option and the process-innovation state are data inputs, while the levels at which those drifts plateau are continuation-value assumptions. The maintained closure provides about five years of headroom after the sample. The horizon is anchored on the cost side: reported wafer prices step up sharply at the 2-nanometer generation, and cost per transistor, the object the process-innovation state proxies, is widely reported as flattening around the end of 2025, so the sample-era improvement trend stops being a defensible expectation past that date.

Within the hazard family fit to the full history, the estimate barely moves. The maintained specification gives a scale of 5.13 with an asymptotic standard error of 0.66; lowering the exposure curvature gives 5.23 and the Poisson family 5.21, and the returns under both stay close to the maintained values of \$3.00 net private, \$8.95 social, and a \$5.95 wedge per R&D dollar. Across the eight hazard specifications estimated on the full history, the three printed here and five further variants (exposure curvature 0.60, the Poisson hazard at exposure curvatures 0.25 and 1.00, and the identity and power exposure forms), the scale ranges from 4.9 to over 5.2, the net private return from about \$2.7 to \$3.0, the social return from about \$8.5 to \$9.0, and the wedge from about \$5.8 to \$6.0. The hazard window is the one primitive that moves the estimate visibly: fitting the hazard to the post-2010 history alone lowers the scale to 3.15, with standard error 0.44, and raises the returns to \$3.53 net private and \$10.69 social, a wedge of \$7.16. Deferring the plateau to seven or ten years after the sample raises the scale slightly, to 5.46 and 5.40. I also estimated a version of the model with firm-specific R&D-cost scales; the data do not identify the firm-specific scales precisely, and I therefore keep the more parsimonious single-scale specification.

F.3 The R&D Allocation Share

The maintained specification allocates each firm's reported R&D to the GPU segment in proportion to the segment's share of revenue (Section 2), which assumes equal R&D spending per dollar of revenue across the firm's businesses. Suppose instead that GPU R&D is k times the allocated series, with k common across firms and over time. To give a rough sense of the sensitivity to this assumption, I rescale accounting R&D while holding every other simulated outcome and the realized public outlays fixed.

The exposure identity links this allocation change to the curvature checks. For the log-concave exposure $g(x; \kappa) = \kappa^{-1} \log(1 + \kappa x)$, $g(kI; \kappa) = k g(I; \kappa k)$. Abstracting from the R&D-intensity bound, fitting the rescaled series at curvature κ therefore corresponds to fitting the original series at curvature κk , adjusting the success-rate parameters, and dividing that specification's cost scale by k . The curvature checks in Appendix F.2 move the estimated cost scale and the industry return little, but they do not re-solve the policy exercises. The calculations below therefore isolate the accounting effect of the allocation and abstract from any equilibrium response to the curvature change.

For the accounting calculation, the R&D-cost scale becomes χ/k . The full-industry social return of s per R&D dollar in Table 3 becomes $(s + 1)/k - 1$: the gross surplus generated by the industry's research is unchanged, but the accounting R&D deducted from it, which is also the denominator, is multiplied by k . Additionality under each instrument is likewise multiplied by k , with the credit's rate divided by k so that its outlay stays fixed. Welfare per public dollar then changes from w to $w - (k - 1)A$, where A is the instrument's additionality in Table 7. Because additionality is small in these experiments, this adjustment is also small.

For illustration, take $k = 1.2$, about the largest common multiplier that allocates no more than each firm's total reported R&D to GPUs.⁵⁰ The R&D-cost scale is then 4.27 rather than 5.13, and the full-industry social return is \$7.29 per R&D dollar rather than \$8.95. As an approximate cross-check, a specification with exposure curvature $\kappa = 0.60$ gives a cost scale of 4.23 and a social return of \$7.34 after conversion, close to the accounting values.

Additionality under the R&D credit rises from 0.13 to 0.16. Welfare per public dollar becomes \$0.49 for the credit, \$0.24 for the premium rebate, and \$1.09 for the non-premium rebate, compared with \$0.51, \$0.23, and \$1.09 in Table 4. Under this accounting adjustment, the credit still ranks first on additionality, while the non-premium rebate ranks first on welfare per public dollar.

⁵⁰The bound is about 1.19, the reciprocal of the largest GPU revenue share in the panel.

G Notation

This appendix collects the symbols used in the model and the appendices, with the unit conventions behind the reported tables. Table [G.1](#) lists the static demand and supply objects first, then the dynamic states and R&D objects, and last the policy and welfare objects; a product j is a firm-tier alternative, and the market index t is quarterly.

Table G.1. Notation for the model and the appendices

Object	Definition
<i>Static demand and supply</i>	
p_{jt}	Price the firm receives for product j (the consumer pays $p_{jt} - b_d$ on a rebate-covered product); the demand estimation and the pricing game carry it in thousands of dollars
q_{jt}	Quality, the PassMark G3D benchmark index in units of 100,000 points
s_{jt}	Inside-product market share
$\mathcal{J}_t, \mathcal{J}_f, \mathcal{J}_f^{\text{int}}$	The quarter's six inside products; firm f 's three-tier menu; its two interior tiers
$\gamma; \bar{\alpha}, \sigma_{\text{price}}; \psi_{f(j)}$	Demand parameters: quality coefficient; mean price coefficient and price-sensitivity dispersion; firm fixed effect
$\xi_{jt}, \varepsilon_{jt}^{\text{MC}}$	Unobserved demand and marginal-cost shocks ($ \varepsilon_{jj} $ instead denotes the mean absolute own-price elasticity)
MC_{jt}	Marginal production cost, in the same units as price
$c_0\text{--}c_5$	Marginal-cost projection coefficients (equation (3.7), Table 2); $c_1 + c_2 t$ forms z_t
π_{jt}	Product variable profit, $M_t s_{jt} (p_{jt} - \text{MC}_{jt})$; the dynamic model carries profit, R&D, and values in millions of dollars
<i>Dynamic states and R&D</i>	
x_t	Dynamic state vector, $x_t = (q_{1t}^{\text{max}}, q_{2t}^{\text{max}}, u_{0t}, z_t, M_t)$
$q_{ft}^{\text{max}}, q_t^{\text{max}}$	Firm f 's quality frontier; the industry frontier, the higher of the two firms' frontier qualities
$z_t, z_{\min}$	Process-innovation state, $z_t = c_1 + c_2 t$, and its long-run floor
$u_{0t}, u_{0,\text{max}}$	Outside-option level and its ceiling (Δ_u is its quarterly drift)
$M_t; \rho_M, M_\infty$	Addressable desktop market size, in millions of units; its persistence and long-run level
χ	R&D-cost scale
ι_{ft}	Firm f 's R&D intensity: spending relative to expected equilibrium stage-game profit $\mathbb{E}[\pi_f^*(x_t)]$
I_{ft}	Accounting GPU-segment R&D, carried in millions of dollars like profit; the hazard's exposure g is evaluated on R&D in hundreds of millions, the scale on which the curvature κ is defined
h_f	Innovation arrival probability, $r_{ft}/(1 + r_{ft})$ with arrival rate r_{ft} ; distinct from the horizon index h
a_f, φ	Firm-specific success-rate intercept and common fishing-out parameter
$g(\cdot); \kappa$	Log-concave R&D exposure in the innovation hazard, $g(x) = \kappa^{-1} \log(1 + \kappa x)$, and its curvature
m_J, v_J	Mean and variance of the log jump size, the frontier advance conditional on success
V_f	Firm f 's value function in the dynamic game (equation (3.8))
β	Quarterly discount factor
<i>Policy and welfare</i>	
W	Within-market surplus, $W = \text{CS} + \Pi - I - G$: consumer surplus plus the firms' profit Π before R&D (operating profit plus the public transfers they receive), less R&D spending I and the public outlay G , which equals those transfers (both zero absent a policy)
B	Common planned present-value budget of the three instruments
G	Realized discounted public outlay of a policy
$\tau; b_d$	R&D-credit rate; rebate per covered unit, with d indexing the rebate design

Notes: Prices and marginal costs enter the demand system and the pricing game in thousands of dollars, and quality enters in units of 100,000 benchmark points; the coefficients of Tables 1 and 2 are on those scales. Converting a margin back to dollars and multiplying by unit sales in millions of units yields profit in millions of dollars, the unit carried by the dynamic model for profit, R&D, and values.